

Fundamental Principal of Counting (FPC)

(i) Sum Rule: If an operation can be performed in m different ways and another can be performed in n different ways and if these two events cannot occur simultaneously. Then one of the two can occur in $m+n$ different ways.

Ex: If there are 14 boys and 12 girls in a class, find the no. of ways of selecting a CR.

Sol No. of Boys = 14
No. of Girls = 12
Total Students in class = $14 + 12 = 26$

∴ there are 26 ways to select a CR. (either Boy or girl)

(ii) Product Rule:— If one operation can be performed in m different ways and another can be performed in n different ways. If the second operation does not depend upon the first then the two occur simultaneously in $m \cdot n$ different ways.

Ex: In a Bench there are six seats vacant. In how many ways can 4 persons seat themselves.

Sol No. of Seats = 6
No. of person = 4

1	2	3	4	5	6
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No. of ways for 1st person to occupy seat = 6

" " " 2nd " " " = 5

" " " 3rd " " " = 4

" " " 4th " " " = 3

∴ By F.P.C Total No. of ways = $6 \times 5 \times 4 \times 3$

= 360.

②

Factorial:- The Continuous product of first n natural numbers is called factorial of ' n ' and is denoted by $n!$ or $\underline{n}$.

$$\therefore \underline{n} = 1.2.3.4. \dots n \quad (\text{Ascending order})$$

$$\underline{n} = n.(n-1)(n-2)(n-3) \dots 3.2.1 \quad (\text{descending order})$$

Note: ① $\underline{0} = 1$.

② $\underline{n} = n(n-1)(n-2)(n-3) \dots 3.2.1$

$$\underline{n} = n \underline{n-1}$$

$$\underline{n} = n(n-1) \underline{n-2}$$

$$\underline{n} = n(n-1)(n-2) \underline{n-3}$$

③ factorial of proper fraction and Negative number is not defined.

Ex ② Compute: $\frac{\underline{52}}{\underline{47.5}} = \frac{52 \cdot \cancel{41} \cdot \cancel{56} \cdot 49 \cdot \cancel{48} \cdot \cancel{47}}{\cancel{47} \cdot \cancel{4} \cdot \cancel{8} \cdot \cancel{7} \cdot 1}$

$$= 52 \times 17 \times 5 \times 49 \times 12$$
$$= 2598960.$$

Ex ② Find ' n ' if $\underline{n+2} = 2550 \underline{n}$.

Sol: $\underline{n+2} = 2550 \underline{n}$

$$\Rightarrow (n+2)(n+1) \cancel{\underline{n}} = 2550 \cancel{\underline{n}}$$

$$\Rightarrow (n+2)(n+1) = 2550$$

$$\Rightarrow (n+2)(n+1) = 51 \times 50 = (49+2)(49+1)$$

$$\Rightarrow \boxed{n=49}$$

$$\textcircled{\text{Ex 2}} \quad \frac{\lfloor 2n \rfloor}{\lfloor n \rfloor} = 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1) \cdot 2^n$$

$$\begin{aligned} \text{Sol:} \quad \frac{\lfloor 2n \rfloor}{\lfloor n \rfloor} &= \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \dots \cdot (2n-1)(2n)}{\lfloor n \rfloor} \\ &= \frac{(1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)) (2 \cdot 4 \cdot 6 \cdot 8 \cdot \dots \cdot 2n)}{\lfloor n \rfloor} \\ &= \frac{(1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)) 2^n (1 \cdot 2 \cdot 3 \cdot 4 \cdot \dots \cdot n)}{\lfloor n \rfloor} \\ &= \frac{(1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)) 2^n \cdot \lfloor n \rfloor}{\lfloor n \rfloor} \\ &= (1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)) 2^n = \text{R.H.S} \end{aligned}$$

$$\textcircled{\text{Ex 2}} \quad \text{find 'n' if } \lfloor n+1 \rfloor = 60 \cdot \lfloor n-1 \rfloor$$

Permutation: → The different Arrangements which can be made out of a given number of things taking some or all of them at a time is called the Permutations.
The Permutation of 'n' different things taking 'r' at a time is denoted by nP_r or $P(n, r)$ and given by

$$nP_r = \frac{n!}{n-r!}$$

$$\text{Ex: } {}^{12}P_5 = \frac{12!}{12-5!} = \frac{12!}{7!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot \cancel{7!}}{\cancel{7!}} = 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = \dots$$

Ex: Find 'n' if $30P(n, 6) = P(n+2, 7)$

Sol: $30P(n, 6) = P(n+2, 7)$

$$\Rightarrow 30 \frac{n!}{n-6!} = \frac{(n+2)!}{(n+2-7)!} \Rightarrow 30 \frac{n!}{n-6!} = \frac{(n+2)!}{n-5!}$$

$$\Rightarrow 30 \frac{\cancel{n!}}{n-6!} = \frac{(n+2)(n+1) \cdot \cancel{n!}}{(n-5) \cdot \cancel{n-6!}}$$

$$\Rightarrow 30 = \frac{(n+2)(n+1)}{(n-5)} \Rightarrow 30(n-5) = (n+2)(n+1)$$

$$\Rightarrow 30n - 150 = n^2 + 3n + 2$$

$$\Rightarrow n^2 - 27n + 152 = 0 \Rightarrow n^2 - 8n - 19n + 152 = 0$$

$$\Rightarrow n(n-8) - 19(n-8) = 0$$

$$\Rightarrow (n-8)(n-19) = 0$$

$$\Rightarrow n-8=0 \quad | \quad n-19=0$$

$$\Rightarrow n=8 \quad | \quad n=19$$

$$\therefore n = 8, 19$$

Practical Problems →

① How many 3-letter words can be formed by using the letters of the word ORIENTAL

Sol: No. of letters in the word ORIENTAL = 8

We want a 3-letter words

∴ Out of 8 letters we are using 3 letters at a time.

∴ Arrangement of 8 letters taking 3 at a time

$$\therefore \text{Required No. of words} = {}^8P_3 = \frac{8!}{8-3!} = \frac{8!}{5!} = \frac{8 \cdot 7 \cdot 6 \cdot 5!}{5!} = 336.$$

② Find the Number of words formed by using the ~~words~~ letters of word TRIANGLE and how many of them

(i) Begin with T (ii) End with E (iii) T and E occupy end places.

Sol: No. of letters in the word TRIANGLE = 8

$$\therefore \text{Total No. of Arrangements} = {}^8P_8 = 8! = 40320.$$

(i) Since Each word Begin with T

T * * * * *

∴ we fix T at the first place

Now we have 7 places and remaining 7 letters

$$\therefore \text{No. of Arrangements of 7 letters taking 7 at a time} = {}^7P_7 = 7! = 5040.$$

$$\therefore \text{Required No. of words} = 1 \times 5040 = 5040.$$

(ii) Since Each word Ends with E

* * * * * E

∴ we fix E at the End place

∴ No. of ways of Arranging E = 1

Now we have remaining 7 letters and also 7 places

$$\therefore \text{No. of ways of filling 7 places with 7 letters} = {}^7P_7 = 7! = 5040$$

$$\therefore \text{By FPC Total No. of words} = 1 \times 5040 = 5040$$

- (iii) Since T and E occupy end places
- $$\begin{array}{c} T - - - - E \\ E - - - - T \end{array}$$
- $\therefore$ No. of ways of Arranging T & E = 2
- Now we have 6 letters and also 6 places
- $\therefore$ No. of ways of filling 6 places with 6 letters = ${}^6P_6 = 16$
 $= 720$
- $\therefore$ By F.P.C Total No. of words = $2 \times 720 = 1440$.

Permutation when the Things are not all different:

- The Number of Permutation / Arrangements of n things taken all at a time when p of them are of one kind q of them are of second kind and r of them are of third kind are given by

$$\text{No. of Permutations} = \frac{n!}{p! q! r!}$$

Ex: - In how many ways the letter of the words MISSISSIPPI be arranged.

Sol: In the word MISSISSIPPI

Total No. of letters = 11

here I occur 4 times

S " 4 times

P " 2 times

$$\therefore \text{Total No. of Arrangements} = \frac{11!}{4! \cdot 4! \cdot 2!} = 34650$$

Ex: Now the word is EXAMINATION.

No. of Letters = 11

A occur 2 time

I " 2 time

N " 2 time

$$\text{Total No. of Permutations} = \frac{11!}{2! \cdot 2! \cdot 2!}$$

Circular Permutations:-

① No. of ways in which n things/Person can be arranged in a Circle = $\underline{n-1}$

② In Case of Necklaces with Beads or Garlands with different flowers.
Total No. of Circular permutations = $\frac{1}{2} \underline{n-1}$

(Ex) In how many ways can 8 things be arranged in
(i) a straight line (ii) in a Circle.

● Sol : No. of things $n = 8$

(i) No. of Arrangement in a st. line = ${}^8P_8 = \underline{8} = 40320$

(ii) No. of Arrangements in a Circle = $\underline{8-1} = \underline{7} = 5040.$

(Ex) In how many ways can 8 beads of different colour form a necklace.

● Sol: No. of Beads, $n = 8$

$$\therefore \text{Required No. of Necklaces} = \frac{1}{2} \underline{n-1} = \frac{1}{2} \underline{8-1} = \frac{\underline{7}}{2} \\ = \frac{5040}{2} = 2520$$

Note :- ① ${}^nP_r = n \cdot {}^{n-1}P_{r-1}$

② ${}^nP_r = (n-r+1) \cdot {}^nP_{r-1}$

③

Combination:- A Combination is a Selections of things taking some or all of them at a time. The Combination of n different things taking r at a time is denoted by nC_r or $C(n, r)$ or $\binom{n}{r}$ and given by

$$\boxed{{}^nC_r = \frac{n!}{(n-r)! \cdot r!}}$$

Note (1) Permutation Means $\rightarrow$ Arrangement $\rightarrow {}^nP_r$
Combination " $\rightarrow$ Selection. $\rightarrow {}^nC_r$

$$(2) \quad {}^nP_r = \frac{n!}{(n-r)!} \quad {}^nC_r = \frac{n!}{(n-r)! r!}$$

$$\therefore {}^nP_r > {}^nC_r$$

$$(3) \quad \boxed{{}^nP_r = r! \cdot {}^nC_r} \quad \text{or} \quad \boxed{{}^nC_r = \frac{{}^nP_r}{r!}}$$

$$(4) \quad {}^nC_r = {}^nC_{n-r}$$

$$(5) \quad {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$$(6) \quad {}^nC_p = {}^nC_q \Rightarrow \text{either } p=q \text{ or } p+q=n$$

Ex 2 In how many ways a team of 11 players be selected out of 15 players, ^{always} how many of them have Two particular players and how many of them do not have two particular players.

Sol Total No. of players = 15

$\therefore$ Now we have to select 11 players out of 15

$$\begin{aligned}\therefore \text{Total No. of Selection} &= {}^{15}C_{11} = \frac{15!}{11! (15-11)!} = \frac{15!}{11! \cdot 4!} \\ &= \frac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 15 \cdot 7 \cdot 13 = 1365\end{aligned}$$

(i) Since Two particular players are always be selected

$\therefore$ No. of ways of selecting that Two particular players = 1

Now out of Remaining 13 players we have to select 9 player

$$\begin{aligned}\therefore \text{No. of ways} &= {}^{13}C_9 = \frac{13!}{9! (13-9)!} = \frac{13!}{9! \cdot 4!} = \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \\ &= 13 \cdot 11 \cdot 5 = 715\end{aligned}$$

(ii) Since Two particular players are never selected

$\therefore$ We Exclude that two particular players from the total No. of players.

$\therefore$ We have to select 11 players out of $15-2=13$ player

$$\begin{aligned}\therefore \text{No. of ways} &= {}^{13}C_{11} = \frac{13!}{11! (13-11)!} = \frac{13!}{11! \cdot 2!} \\ &= \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 13 \cdot 6 = 78\end{aligned}$$

Pigeonhole Principle:— If n pigeons are assigned m pigeonholes, where n is sufficiently large as compared to m then One of the pigeonholes must contain at least $\left(\frac{n-1}{m}\right) + 1$ pigeons.

PERMUTATION AND COMBINATION

Art-1. Factorial Notations

The product of all the positive integers from 1 to n is called *factorial n* and is denoted by the symbol $\underline{n}$ or $n!$. For example,

$$\underline{6} = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6$$

and $1 \cdot 2 \cdot 3 \cdot 4 \dots 20 = \underline{20}$

An Important result

$$\underline{n} = n(n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1$$

$$= n[(n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1] = n\underline{n-1}$$

$$= n(n-1)\underline{n-2} = n(n-1)(n-2)\underline{n-3}$$

.....

$$\text{Also } n\underline{n-1} = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1$$

$$\therefore n\underline{n-1} = \underline{n}$$

$$\text{Similarly } (n+2)(n+1)\underline{n} = \underline{n+2}$$

Note 1. Since $\underline{n} = n\underline{n-1}$

$$\text{Put } n = 1$$

$$\therefore \underline{1} = 1\underline{0}$$

$$\therefore 1 = (1)(\underline{0})$$

$$\Rightarrow \underline{0} = 1$$

We will also prove at a later stage that $\underline{0} = 1$

Note 2. Factorial of proper fraction and negative integer is not defined.

Example 1. Compute $\frac{\lfloor 52 \rfloor}{\lfloor 47 \rfloor \lfloor 5 \rfloor}$.

Sol. $\frac{\lfloor 52 \rfloor}{\lfloor 47 \rfloor \lfloor 5 \rfloor} = \frac{52 \times 51 \times 50 \times 49 \times 48 \times \lfloor 47 \rfloor}{\lfloor 47 \rfloor (5 \times 4 \times 3 \times 2 \times 1)} = 2598960$

Example 2. If $\frac{1}{\lfloor 9 \rfloor} + \frac{1}{\lfloor 10 \rfloor} = \frac{x}{\lfloor 11 \rfloor}$; find x .

Sol. $\frac{1}{\lfloor 9 \rfloor} + \frac{1}{\lfloor 10 \rfloor} = \frac{x}{\lfloor 11 \rfloor}$

$$\Rightarrow \frac{1}{\lfloor 9 \rfloor} + \frac{1}{10 \times \lfloor 9 \rfloor} = \frac{x}{11 \times 10 \times \lfloor 9 \rfloor}$$

$$\Rightarrow \frac{1}{1} + \frac{1}{10} = \frac{x}{110} \Rightarrow 110 + 11 = x$$

$$\Rightarrow x = 121$$

Example 3. If $\lfloor n+2 \rfloor = 2550 \lfloor n \rfloor$, find n .

Sol. $\lfloor n+2 \rfloor = 2250 \lfloor n \rfloor$

$$\Rightarrow (n+2)(n+1) \lfloor n \rfloor = 2550 \lfloor n \rfloor \Rightarrow (n+2)(n+1) = 2550$$

$$\Rightarrow n^2 + 3n - 2548 = 0$$

$$\Rightarrow n = \frac{-3 \pm \sqrt{9 + 10192}}{2} = \frac{-3 \pm \sqrt{10201}}{2} = \frac{-3 \pm 101}{2}$$

$$\Rightarrow n = 49, -52$$

Rejecting $n = -52$ as n cannot be negative, we get $n = 49$.

Example 4. If $\frac{\lfloor n \rfloor}{2 \lfloor n-2 \rfloor}$ and $\frac{\lfloor n \rfloor}{4 \lfloor n-4 \rfloor}$ are in the ratio 2 : 1, find the value of n .

Sol. Here $\frac{\lfloor n \rfloor}{2 \lfloor n-2 \rfloor} : \frac{\lfloor n \rfloor}{4 \lfloor n-4 \rfloor} = 2 : 1$

$$\Rightarrow \frac{\lfloor n \rfloor}{2 \lfloor n-2 \rfloor} \times \frac{4 \lfloor n-4 \rfloor}{\lfloor n \rfloor} = \frac{2}{1} \Rightarrow \frac{4 \lfloor n-4 \rfloor}{2 \lfloor n-2 \rfloor} = \frac{2}{1}$$

$$\Rightarrow \frac{4 \cdot 3 \cdot 2 \cdot 1 \lfloor n-4 \rfloor}{2(n-2)(n-3) \lfloor n-4 \rfloor} = \frac{2}{1} \Rightarrow \frac{6}{(n-2)(n-3)} = \frac{1}{1}$$

$$\Rightarrow (n-2)(n-3) = 6 \Rightarrow (n-2)(n-3) = (3)(2)$$

$$\Rightarrow n-2 = 2, \text{ or } n = 5$$

Example 5. Prove that $\frac{|2n}{|n} = 1 \cdot 3 \cdot 5 \dots (2n-1) \cdot 2^n$

$$\begin{aligned} \text{Sol. L.H.S.} &= \frac{|2n}{|n} \\ &= \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \dots (2n-1)(2n)}{|n} = \frac{[1 \cdot 3 \cdot 5 \dots (2n-1)][2 \cdot 4 \cdot 6 \dots (2n)]}{|n} \\ &= \frac{[1 \cdot 3 \cdot 5 \dots (2n-1)] \cdot 2^n [1 \cdot 2 \cdot 3 \dots n]}{|n} \\ &= \frac{[1 \cdot 3 \cdot 5 \dots (2n-1)] \cdot 2^n |n}{|n} = 1 \cdot 3 \cdot 5 \dots (2n-1) 2^n \\ &= \text{R.H.S.} \end{aligned}$$

Example 6. If largest of 87 consecutive integers is 326, what is smallest?

(B.C.A.-II 2008)

Sol. Number of integers = 87

Largest integer = 326

Smallest integer = $326 - 87 + 1 = 240$

EXERCISE 3 (a)

1. Evaluate

(i) $|4$ (ii) $|6$ (iii) $|7$ (iv) $|8 - |5$

(v) $|6 - |5$ (vi) $2|6 - 3|5$ (vii) $3|4 + 7|4$

2. Compute $|2 + |3$. Is $|2 + |3 = |5$?

3. Compute $|4 |2$. Is $|4 |2 = |8$?

4. Find $\frac{|6}{|5}$.

5. Compute $\frac{|8}{|4}$. Is $\frac{|8}{|4} = |2$?

6. Compute $\frac{|20}{|18 |20-18}$.

7. Evaluate (i) $\frac{\lfloor 6 \rfloor}{2 \times \lfloor 4 \rfloor}$ (ii) $\frac{\lfloor 7 \rfloor}{\lfloor 4 \times \lfloor 2 \rfloor}$

8. Evaluate $\lfloor n-r \rfloor$ when

(i) $n = 6, r = 2$ (ii) $n = 9, r = 5$

9. Evaluate $\frac{\lfloor n \rfloor}{\lfloor n-r \rfloor}$ when

(i) $n = 10, r = 4$ (ii) $n = 12, r = 3$ (iii) $r = 1$
(iv) $r = 2$ (v) $r = 3$

10. Evaluate $\frac{\lfloor n \rfloor}{\lfloor r \rfloor \lfloor n-r \rfloor}$, when

(i) $n = 6, r = 2$, (ii) $n = 7, r = 4$ (iii) $n = 15, r = 12$

11. Prove that $\lfloor n \rfloor \cdot (n+2) = \lfloor n \rfloor + \lfloor n+1 \rfloor$.

12. If $\lfloor n+1 \rfloor = 60 \lfloor n-1 \rfloor$, find n .

13. Prove that $\lfloor n \rfloor + 1$ is not divisible by any integer between 2 and n .

ANSWERS

- | | | |
|----------------|-----------------------|---------------------|
| 1. (i) 24 | (ii) 720 | (iii) 5040 |
| (iv) 40200 | (v) 600 | (vi) 1080 (vii) 240 |
| 2. 8 ; no | 3. 48 ; no | 4. 6 |
| 5. 1680 ; no | 6. 190 | |
| 7. (i) 15 | (ii) 105 | |
| 8. (i) 24 | (ii) 24 | |
| 9. (i) 5040 | (ii) 1320 | (iii) n |
| (iv) $n^2 - n$ | (v) $n^3 - 3n^2 + 2n$ | |
| 10. (i) 15 | (ii) 35 | (iii) 455 |
| 12. no value | | |

Art-2. Fundamental Principle of Counting i.e. FPC

If one operation can be performed in ' m ' different ways and if corresponding of these m ways of performing the first operation, there are ' n ' different ways of performing the second operation, then the number of different ways of performing two operations taken together is $m \times n$.

Extension. If corresponding to each of the $m \times n$ ways of performing the two operations taken together, there are ' p ' different ways of performing the third operation then the number of different ways of performing the three operations taken together is $m \times n \times p$ and so on.

Note. Fundamental Principle of Counting is known as Fundamental Principle of Association or Multiplication Principle.

We give some example to illustrate the above principle.

ILLUSTRATIVE EXAMPLES

Example 1. Ram proposes to go to his friend Ali's house in a town which is connected with his village by three different routes. From there he will go to the city where his uncle lives. The city is connected with the town by two different routes. List out the various possible routes which Ram can choose from his village to go to the city.

Sol. Let us denote the three different routes connecting the village to the town by R_1, R_2, R_3 . To go to the town, Ram can use any route R_1, R_2 or R_3 . Thus, there are three choices of routes for Ram to go to the town. Then, for each one of these routes, he can go to the city from the town by any one of the two different routes, say either by r_1 or r_2 . The various possible choices of routes for Ram to go to the city are $R_1 r_1, R_1 r_2, R_2 r_1, R_2 r_2, R_3 r_1, R_3 r_2$.

$$\therefore \text{total number of possible routes} = 3 \times 2 = 6.$$

Example 2. If there are 20 steamers playing between places A and B, in how many ways could the round trip from A be made if the return was made on (i) the same steamer, (ii) a different steamer?

Sol. (i) Number of steamers = 20

Since the return is made by the same steamer.

$$\therefore \text{total number of ways} = 20$$

(ii) Number of ways from A to B = 20

$$\text{Number of ways from B to A} = 19$$

[$\because$ Journey is to be made by different steamer from B to A]

$$\therefore \text{total number of ways} = 20 \times 19 = 380$$

Example 3. How many 4-letter code words are possible using the first 10 letters of the English alphabet, if no letter can be repeated?

Sol. Number of letters = 10

First letter of the code can be selected in 10 ways

Second letter of the code can be selected in 9 ways

Third letter of the code can be selected in 8 ways

Fourth letter of the code can be selected in 7 ways

$\therefore$ by fundamental principle of counting,

$$\text{total number of 4-letter code words} = 10 \times 9 \times 8 \times 7 = 5040$$

Example 4. How many 5-digit telephone numbers can be constructed using the digits 0 to 9 if each number starts with 67, for example 67125 etc., and no digit appears more than once?

Sol. Number of digits from 0 to 9 = 10

We are to fill up five places

6 is in 1st place and 7 is in 2nd place.

X	X	X	X	X	X
6	7				

Third place can be filled up in 8 ways as any of the remaining 8 digits can be placed there.

Similarly fourth and fifth places can be filled in 7 and 6 ways

$\therefore$ by fundamental principle of counting,

total number of telephone numbers = $8 \times 7 \times 6 = 336$

Example 5. A class consists of 40 girls and 60 boys. In how many ways can a president, vice president, treasurer and secretary be chosen if the treasurer must be a girl and a secretary must be a boy and a student may not hold more than one office?

Sol. Number of boys = 60

Number of girls = 40

Since treasurer is a girl

$\therefore$ treasurer can be selected in 40 ways

Again secretary is a boy

$\therefore$ secretary can be selected in 60 ways.

After the two selections are made,

remaining two selections can be made in 98×97 ways

$\therefore$ total number of ways = $60 \times 40 \times 98 \times 97 = 22814400$

Example 6. Numbers 1, 2 and 3 are written on three cards. How many 2-digit numbers can be formed by placing two cards side by side?

Sol. Given digits are 1, 2, 3 and two digit numbers are to be formed.

Ten's place can be filled up in 3 different ways as one of the cards can be placed there. After filling ten's place, there are two different ways of filling up the unit's place as any one of the remaining two cards can be placed there.

$\therefore$ required numbers = $3 \times 2 = 6$.

Example 7. How many numbers can be formed from the digits 1, 2, 3, 9 if repetition of digits is not allowed?

Sol. (a) Numbers with one digit : There are four digits, hence four numbers of one digit can be formed with the help of these digits.

Hence, number of one digit numbers = 4.

(b) Numbers with two digits : First place of two digit number can be filled in 4 ways and the second place can be filled in 3 ways.

Hence, number of two digit numbers = $4 \times 3 = 12$.

(c) Numbers with three digits :

Number of three digits number = $4 \times 3 \times 2 = 24$.

(d) Number with four digits :

Number of four digits numbers = $4 \times 3 \times 2 \times 1 = 24$.

Hence, total number of digits formed with the given digits = $4 + 12 + 24 + 24 = 64$.

Example 8. Find the number of different signals that can be generated by arranging at least two flags in order (one below the other) on a vertical staff, if five different flags are available.

Sol. Number of flags = 5

A signal can be formed by using two, three, four or five flags.

Number of ways of forming signal using two flags = $5 \times 4 = 20$

Number of ways of forming signal using three flags = $5 \times 4 \times 3 = 60$

Number of ways of forming signal using four flags = $5 \times 4 \times 3 \times 2 = 120$

Number of ways of forming signal using five flags = $5 \times 4 \times 3 \times 2 \times 1 = 120$

$\therefore$ total numbers of signals formed = $20 + 60 + 120 + 120 = 320$.

Example 9. Let repetition of digits in number is allowed. How may 4 digit numbers can be formed using digits 1, 2, 3, 5, 7, 8. How many of them are

(a) Even (b) Multiplicative of 5 (c) Less than 4000.

(P.T.U. B.C.A. I 2007)

Sol. Number of 4 digits number formed using 1, 2, 3, 5, 7, 8

$$= 6 \times 6 \times 6 \times 6 = 1296$$

(a) For a number to be even unit digit must be even i.e. 2 or 8

Total even numbers = $6 \times 6 \times 6 \times 2 = 432$

(b) For a number to multiplicative of 5 unit digit must be 5

Such numbers are = $6 \times 6 \times 6 \times 1 = 216$

(c) A number is less than 4000 if digit at 1000 place is 1 or 2 or 3.

Number less than 4000 = $3 \times 6 \times 6 \times 6 = 648$

Example 10. How many integers between 10^5 and 10^6 have digits other than 0, 2, 5 or 8 ?

(P.T.U. B.C.A. I 2006)

Sol. Digits other than 0, 2, 5, 8 are 1, 3, 4, 6, 7, 9

Between 10^5 (100000) and 10^6 (1000000) these are only 6 digits numbers.

Numbers formed using 1, 3, 4, 6, 7, 9 = $6 \times 6 \times 6 \times 6 \times 6 \times 6 = 6^6$.

Example 11. A variable name in programming language must be either a letter or letter followed by decimal digit. How many variable name are possible.

(P.T.U. B.C.A. I 2007)

Sol. Two case arise :

(i) Variable name is letter

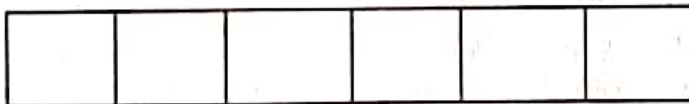
(ii) Variable name is letter followed by digit.

Number of variables formed = $26 + 260 = 286$.

EXERCISE 3 (b)

1. There are 5 routes from place A to place B and 3 routes from place B to place C. Find how many different routes are there from A to C via B.
2. Let there be three jobs a, b, c and three persons A, B and C. Find the possible number of assignments of jobs so that one person is assigned only one job.
3. John wants to go abroad by ship and return by air. He has a choice of 6 different ships to go and 4 airlines to return. In how many ways can he perform his journey?
4. In how many ways can 5 persons draw water from 5 taps, assuming no tap remains unused?
5. In how many ways can 3 people be seated in a row containing 7 seats?
6. A coin is tossed twice and the outcomes are recorded. Find the number of possible outcomes.
7. In a class there are 30 boys and 18 girls. The teacher wants to select one boy and one girl to represent the class for a quiz competition. In how many ways can the teacher make this selection?
8. Six pictures are to be arranged from left to right on a wall of an art gallery for display. How many arrangements are possible?
9. The digits, from 0 to 9, are written on slips of paper and placed in a box. Three of the slips of paper are drawn and placed in order. How many different outcomes are possible?
10. For a group photograph, 3 boys and 2 girls stand in a line in all possible ways. How many photos could be taken if each photo corresponds to each such arrangement?
11. A sample of 3 bulbs is tested. A bulb is labelled as 'G' if it is good and 'D' if it is defective. Find the number of all the possible outcomes.
12. Kartar goes to a movie. The cinema hall has two entrances and three exits. In how many ways can Kartar enter and exit from the hall?
13. There are 4 multiple choice questions in an examination. How many sequences of answers are possible, if each question has 2 choices?
14. How many 3-letter code words are possible using the first 10 letters of English alphabet if
 - (i) no letter can be repeated?
 - (ii) letters are repeated?

15. In how many ways can 5 persons sit in a car, 2 including the driver in the front seat and 3 in the back seat, if 2 particular persons out of the 5 are to avoid the driver's seat?
16. A coin is tossed three times and the outcomes are recorded. How many possible outcomes are there? How many possible outcomes if the coin is tossed four times? Five times? n times?
17. Eight children are to be seated on a bench.
 - (i) In how many ways can the children be seated?
 - (ii) How many arrangements are possible if the youngest child sits at the left hand end of the bench?
18. How many 2-digit even numbers can be formed from the digits 1, 2, 3, 4 and 5 if the digits can be repeated?
19. Given 4 flags of different colours, how many different signals can be generated if the signal requires the use of two flags, one below the other?
20. Find the number of different signals that can be made by arranging at least three flags in order on a vertical pole, if 6 different flags are available.
21. Each of the six squares in strip shown in figure is to be coloured with any one of ten different colours so that no two adjacent squares have the same colour. Find the number of ways of colouring the strip.



ANSWERS

1. 15	2. 6	3. 24	4. 120
5. 210	6. 4	7. 540	8. 720
9. 720	10. 120	11. 8	12. 6
13. 16	14. (i) 720	(ii) 1000	15. 72
16. $2^3, 2^4, 2^5, 2^n$	17. (i) 40320	(ii) 5040	
18. 10	19. 12	20. 1920	21. 590490

Art-3. Permutation

It is an arrangement that can be made by taking some or all of a number of given things.

Meaning of nP_r : nP_r means the number of permutations of n different things taken r at a time.

Illustrations. Consider three letters a, b, c .

(i) The permutations of three letters taken two at a time are :

ab, ba, ac, ca, bc, cb

$\therefore$ The number of arrangements of three letters taken two at a time is 6 i.e., ${}^3P_2 = 6$

Note. nP_r is also written as $P(n, r)$.

Art-4. Combination

It is a group (or selection) that can be made by taking some or all of a number of given things at a time.

Meaning of nC_r : nC_r means the number of combinations of n different things taken r at a time.

Illustration. Consider three letters a, b, c .

The groups of these 3 letters taken two at a time are ab, bc, ca .

As far as group is concerned ac or ca is the same group, as in a group we are concerned with the number of things contained, whereas in the case of arrangement we have to take into consideration the order of things.

Note. nC_r is also written as $C(n, r)$.

Art-5. Find the number of permutation of n different things taken r at a time i.e., find the value of nP_r .

Proof: We know that the number of permutations of n different things taken r at a time is the same as the number of ways in which r places in a line can be filled up with persons.

The first place can be filled up in n different ways as any one of the n persons can be placed there. After filling up the first place in any one of n ways, there are $(n - 1)$ different ways of filling up the second place, as any one of the remaining $(n - 1)$ persons can be placed there. Therefore, by the principle of association, the two places taken together can be filled up in $n(n - 1)$ ways. After the two places have been filled up in any one of the $n(n - 1)$ ways, the third place can be filled up in $(n - 2)$ ways as any one of the remaining $(n - 2)$ persons can be placed there. Therefore the three places taken together can be filled up in $n(n - 1)(n - 2)$ ways.

Proceeding in this way, we see that

(i) Whenever a place is filled up, a new factor is introduced.

(ii) The factors begin with n and go on diminishing by unity.

$$\therefore r\text{th factor} = n - (r - 1) = n - r + 1$$

$\therefore$ number of ways of filling up r places

$$= n(n - 1)(n - 2) \dots (n - r + 1)$$

$$\therefore {}^nP_r = n(n - 1)(n - 2) \dots (n - r + 1)$$

$$= \frac{n(n - 1)(n - 2) \dots (n - r + 1)(n - r) \dots 3 \cdot 2 \cdot 1}{(n - r) \dots 3 \cdot 2 \cdot 1}$$

$$\therefore {}^nP_r = \frac{n!}{(n - r)!}$$

Cor. We know that ${}^nP_r = n(n-1)(n-2)\dots(n-r+1)$... (1)

$$\text{and } {}^nP_r = \frac{{}_nP_n}{{}_nr} \quad \dots (2)$$

Putting $r = n$ in (1) and (2),

$$\begin{aligned} {}^nP_n &= n(n-1)(n-2)\dots(n-n+1) \\ &= n(n-1)(n-2)\dots 1 = {}_n1 \end{aligned} \quad \dots (3)$$

$$\text{and } {}^nP_n = \frac{{}_nP_n}{{}_n0} \quad \dots (4)$$

From (3) and (4), we get,

$${}_n1 = \frac{{}_nP_n}{{}_n0} \Rightarrow {}_n0 = 1.$$

ILLUSTRATIVE EXAMPLES

Example 1. Prove that

$$(i) \quad {}^nP_n = 2 \cdot {}^nP_{n-2} \quad (ii) \quad {}^{10}P_3 = {}^9P_3 + 3 \times {}^9P_2$$

Sol. (i) L.H.S. = ${}^nP_n = {}_n1$

$$\text{R.H.S.} = 2 \cdot {}^nP_{n-2} = 2 \times \frac{{}_nP_n}{{}_n(n-2)} = 2 \times \frac{{}_nP_n}{2} = 2 \times \frac{{}_nP_n}{2 \times 1} = {}_n1$$

$$\therefore \text{L.H.S.} = \text{R.H.S.}$$

$$(ii) \text{ L.H.S.} = {}^{10}P_3 = 10 \times 9 \times 8 = 720$$

$$\begin{aligned} \text{R.H.S.} &= {}^9P_3 + 3 \cdot {}^9P_2 \\ &= 9 \times 8 \times 7 + 3(9 \times 8) = 504 + 216 = 720 \end{aligned}$$

$$\therefore \text{L.H.S.} = \text{R.H.S.}$$

Example 2. Find n if ${}^{30}P(n, 6) = P(n+2, 7)$.

Sol. ${}^{30}P(n, 6) = P(n+2, 7)$

$$\Rightarrow 30 \left[\frac{{}_nP_n}{{}_n(n-6)} \right] = \frac{{}_{n+2}P_{n+2}}{{}_{n+2}(n+2)-7} \Rightarrow \frac{30 {}_nP_n}{{}_n(n-6)} = \frac{(n+2)(n+1) {}_nP_n}{(n-5) {}_n(n-6)}$$

$$\Rightarrow \frac{30}{1} = \frac{(n+2)(n+1)}{n-5}$$

$$\Rightarrow (n+2)(n+1) = 30(n-5)$$

$$\Rightarrow n^2 - 27n + 152 = 0 \Rightarrow (n-8)(n-19) = 0$$

$$\Rightarrow n = 8, 19.$$

Example 3. Find the value of n such that

$$(i) \quad {}^n P_5 = 42 {}^n P_3, \quad n > 4.$$

$$(ii) \quad \frac{{}^n P_4}{{}^{n-1} P_4} = \frac{5}{4}, \quad n > 4.$$

Sol. (i) Given that

$${}^n P_5 = 42 {}^n P_3, \text{ i.e. } n(n-1)(n-2)(n-3)(n-4) = 42 n(n-1)(n-2)$$

As $n > 4$, $n(n-1)(n-2) \neq 0$.

Therefore dividing both sides by $n(n-1)(n-2)$, we get

$$(n-3)(n-4) = 42$$

$$\therefore n^2 - 7n + 12 = 42$$

$$\text{or } n^2 - 7n - 30 = 0$$

$$\text{or } (n-10)(n+3) = 0$$

$$\text{Hence } n = 10, -3$$

Now n cannot be negative

$$\therefore n = 10$$

(ii) Given that

$$\frac{{}^n P_4}{{}^{n-1} P_4} = \frac{5}{3}$$

$$\text{i.e. } 3n(n-1)(n-2)(n-3) = 5(n-1)(n-2)(n-3)(n-4)$$

As $n > 4$, $(n-1)(n-2)(n-3) \neq 0$.

Therefore dividing both sides by $(n-1)(n-2)(n-3)$, we get,

$$3n = 5(n-4) \text{ or } 3n = 5n - 20 \Rightarrow 2n = 20 \Rightarrow n = 10$$

Example 4. If $P(n-1, 3) : P(n, 4) = 1 : 9$, find n .

Sol. Here $P(n-1, 3) : P(n, 4) = 1 : 9$

$$\therefore {}^{n-1} P_3 : {}^n P_4 = 1 : 9$$

$$\Rightarrow \frac{{}^{n-1} P_3}{{}^n P_4} = \frac{1}{9}$$

$$\Rightarrow \frac{(n-1)(n-2)(n-3)}{n(n-1)(n-2)(n-3)} = \frac{1}{9} \Rightarrow \frac{1}{n} = \frac{1}{9} \quad [\because n \neq 1]$$

$$\Rightarrow n = 9.$$

Example 5. Find r , if ${}^{10} P_r = 2 {}^9 P_r$.

Sol. ${}^{10}P_r = 2 \times {}^9P_r$

$$\Rightarrow \frac{|10|}{|10-r|} = 2 \times \frac{|9|}{|9-r|}$$

$$\Rightarrow \frac{10|9|}{(10-r)|9-r|} = 2 \times \frac{|9|}{|9-r|}$$

$$\Rightarrow \frac{5}{10-r} = 1 \Rightarrow 5 = 10-r \Rightarrow r = 5$$

Example 6. Prove that ${}^nP_r = {}^{n-1}P_r + r {}^{n-1}P_{r-1}$.

Sol. R.H.S. $= {}^{n-1}P_r + r {}^{n-1}P_{r-1}$

$$= \frac{|n-1|}{|(n-1)-r|} + r \frac{|n-1|}{|(n-1)-(r-1)|}$$

$$= |n-1| \left[\frac{1}{|n-r-1|} + \frac{r}{|n-r|} \right] = |n-1| \left[\frac{1}{|n-r-1|} + \frac{r}{(n-r)|n-r-1|} \right]$$

$$= \frac{|n-1|}{|n-r+1|} \left[\frac{1}{1} + \frac{r}{n-r} \right] = \frac{|n-1|}{|n-r-1|} \left[\frac{n-r+r}{n-r} \right]$$

$$= \frac{|n-1|}{|n-r-1|} \left[\frac{n}{n-r} \right] = \frac{n|n-1|}{(n-r)|n-r-1|} = \frac{|n|}{|n-r|} = {}^nP_r$$

= L.H.S.

Example 7. If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$, find r .

Sol. ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$

$$\Rightarrow \frac{{}^{56}P_{r+6}}{{}^{54}P_{r+3}} = \frac{30800}{1}$$

$$\Rightarrow \frac{\frac{|56|}{|56-r-6|}}{\frac{|54|}{|54-r-3|}} = \frac{30800}{1} \Rightarrow \frac{|56|}{|50-r|} \times \frac{|51-r|}{|54|} = \frac{30800}{1}$$

$$\Rightarrow \frac{(56)(55)|54|}{|54|} \times \frac{(51-r)|50-r|}{|50-r|} = \frac{30800}{1}$$

$$\Rightarrow 51-r = 10 \Rightarrow r = 41$$

Example 8. Show that for all integers $n \geq 2$

$${}^n P_2 + {}^n P_1 = n^2.$$

Sol. Since $n \geq 2$

$\therefore {}^n P_2$ and ${}^n P_1$ are defined

$${}^n P_2 = \frac{|n|}{|n-2|} = \frac{n(n-1)|n-2|}{|n-2|} = n(n-1)$$

$${}^n P_1 = \frac{|n|}{|n-1|} = \frac{n|n-1|}{|n-1|} = n$$

$$\begin{aligned} \text{Now L.H.S.} &= {}^n P_2 + {}^n P_1 = n(n-1) + n = n^2 - n + n = n^2 \\ &= \text{R.H.S.} \end{aligned}$$

EXERCISE 3 (c)

- Find n if $P(n, 4) = 20 P(n, 2)$.
- Find n if $P(2n, 3) = 100 P(n, 2)$.
- Find n if ${}^{n-1}P_3 : {}^n P_4 = 1 : 9$.
- Find r if
 - ${}^5 P_r = 2 {}^6 P_{r-1}$
 - ${}^5 P_r = 6 {}^5 P_{r-1}$
 - ${}^5 P_r = {}^6 P_{r-1}$
- Prove that if $r \leq s \leq n$, then $P(n, s)$ is divisible by $P(n, r)$.
- Find n if $P(9, 5) + 5 P(9, 4) = P(10, n)$.
- Find r if $P(10, r+1) : P(11, r) = 30 : 11$.

ANSWERS

- | | | |
|------------|---------------|---------|
| 1. 7 | 2. 13 | 3. 9 |
| 4. (i) 3 | (ii) No value | (iii) 4 |
| 6. $n = 5$ | 7. 5 | |

Art-6. Practical Problems involving Permutation

Now we will apply the formula

$$P(n, r) = n(n-1)(n-2) \dots (n-r+1) = \frac{|n|}{|n-r|} \text{ to practical problems.}$$

ILLUSTRATIVE EXAMPLES

Example 1. How many 3-letter words can be made using the letters of the words ORIENTAL ?

Sol. Given word is ORIENTAL

$\therefore$ number of letters = 8

Number of letters to be taken at a time = 3

$\therefore$ required number of words $= {}^8P_3 = 8 \times 7 \times 6 = 336$

Example 2. Six candidates are called for interview to fill four posts in an office. Assuming that each candidate is fit for each post, determine the number of ways in which

(i) first and second posts can be filled.

(ii) first three posts can be filled.

(iii) all the four posts can be filled.

Sol. Number of candidates = 6

(i) Number of posts to be filled = 2

$\therefore$ required number of ways $= {}^6P_2 = 6 \times 5 = 30$

(ii) Number of posts to be filled = 3

$\therefore$ required number of ways $= {}^6P_3 = 6 \times 5 \times 4 = 120$

(iii) Number of posts to be filled = 4

$\therefore$ required number of ways $= {}^6P_4 = 6 \times 5 \times 4 \times 3 = 360$

Example 3. Find the number of different 8-letter words formed from the letters of the word TRIANGLE if each word is to

(i) begin with T (ii) end with E (iii) begin with T and end with E.

Sol. The given word is TRIANGLE

Number of letters = 8

Number of letter to be taken at a time = 8

(i) Since each word is to begin with T

$\therefore$ fix T in the beginning

$\therefore$ 7 letters are to be arranged in 7 places

$\therefore$ required numbers of words $= {}^7P_7 = \underline{7} = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$

(ii) Since each word is to end with E

$\therefore$ fix E in the end

$\therefore$ 7 letters are to be arranged in 7 places

$\therefore$ required number of words $= {}^7P_7 = \underline{7} = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$

(iii) Since each word is to begin with T and end with E

$\therefore$ fix T in the beginning and E in the end

$\therefore$ 6 letters are to be arranged in 6 places

$\therefore$ required number of words $= {}^6P_6 = \underline{6} = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

Example 4. How many words (with or without dictionary meaning) can be made from the letters in the word MONDAY, assuming that no letter is repeated, if

(i) 4 letters are used at a time ?

(ii) all letters used at a time ?

(iii) all letters are used but the first is a vowel ?

Sol. The given word is MONDAY.

Number of letters = 6

(i) Number of letters to be taken at a time = 4

$\therefore$ required number of words $= {}^6P_4 = 6 \times 5 \times 4 \times 3 = 360$

(ii) Number of letters to be taken at a time = 6

$\therefore$ required number of words $= {}^6P_6 = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

(iii) First place with 2 vowels O and A can be filled in 2 ways.

Now remaining 5 places can be filled up with 5 letters in 5P_5 ways.

$\therefore$ required number of words $= 2 \times {}^5P_5$

$$= 2 \times (5 \times 4 \times 3 \times 2 \times 1) = 2 \times 120 = 240$$

Example 5. It is required to seat 5 men and 4 women in a row so that the women occupy the even places. How many such arrangements are possible ?

Sol. Number of men = 5

Number of women = 4

Nine places are to be filled. In these 9 places, only four are even. Therefore, 4 places can be filled with 4 women in ${}^4P_4 = \underline{4}$ ways. Also remaining 5 places can be filled up with 5 men in ${}^5P_5 = \underline{5}$ ways.

$\therefore$ required number of arrangements $= \underline{4} \times \underline{5}$

$$= (4 \times 3 \times 2 \times 1) \times (5 \times 4 \times 3 \times 2 \times 1)$$

$$= 24 \times 120 = 2880$$

Example 6. Find the number of different 8-letter arrangements that can be made from the letters of the word DAUGHTER so that all vowels never occur together.

Sol. Given word is DAUGHTER

Number of letters = 8

Number of letters to be taken at a time = 8

$$\therefore \text{total number of words formed} = {}^8P_8 = \underline{8}$$

$$= 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$$

Let us find the number of words in which three vowels A, U, E, are always together.
Consider the three vowel A, U, E. as one letter.

$$\therefore 6 \text{ letters can be arranged in } \underline{6} \text{ ways.}$$

Also 3 vowels can be arranged among themselves in $\underline{3}$ ways.

$$\therefore \text{total number of words} = \underline{6} \times \underline{3}$$

$$= (6 \times 5 \times 4 \times 3 \times 2 \times 1) \times (3 \times 2 \times 1)$$

$$= 720 \times 6 = 4320$$

$$\therefore \text{required number of words} = 40320 - 4320 = 36000$$

Example 7. In how many ways 5 boys and 3 girls be seated in a row so that no two girls are together?

Sol. Let 5 boys be B_1, B_2, B_3, B_4, B_5

$$\times B_1 \times B_2 \times B_3 \times B_4 \times B_5 \times$$

$\therefore$ no two girls are together

$\therefore$ 3 girls can be arranged in 6 'X' marked places in 6P_3 ways.

Also 5 boys can be arranged among themselves in $\underline{5}$ ways.

$$\therefore \text{required number of ways} = {}^6P_3 \times \underline{5}$$

$$= (6 \times 5 \times 4) \times (5 \times 4 \times 3 \times 2 \times 1)$$

$$= 120 \times 120 = 14400$$

Example 8. Find the number of different 8 letter words formed from the letters of the word TRIANGLE if each word is to

- (i) have no two vowel together
- (ii) have both consonants and vowels together
- (iii) have the relative position of the vowels and consonants unaltered.

Sol. The given word is TRIANGLE.

$$(i) \times T \times R \times N \times G \times L \times$$

$\therefore$ no two vowels are together

$\therefore$ vowels can occupy $\times$ marked places.

$\therefore$ three vowels I, A, E can be arranged in six $\times$ marked places in 6P_3 ways.

Also the five consonants can be arranged among themselves in $\underline{5}$ ways.

$$\therefore \text{the required number of words} = {}^6P_3 \times \underline{5}$$

$$= (6 \times 5 \times 4) \times (5 \times 4 \times 3 \times 2 \times 1)$$

$$= 120 \times 120 = 14400.$$

(ii) Since consonants T, R, N, G, L and vowels I, A, E are both always together
 $\therefore$ consider the five consonants as one letter, also the three vowels as one letter.
 Now the two letters can be arranged in $\underline{2}$ ways.

Also the five consonants can be arranged among themselves in $\underline{5}$ ways and vowels can be arranged in $\underline{3}$ ways.

$$\begin{aligned}\therefore \text{required number of words} &= \underline{2} \times \underline{5} \times \underline{3} \\ &= (2 \times 1) \times (5 \times 4 \times 3 \times 2 \times 1) \times (3 \times 2 \times 1) \\ &= 2 \times 120 \times 6 = 1440.\end{aligned}$$

(iii) The relative position of the vowels and consonants remains unaltered that a vowel can take the place of another vowel and a consonant can take the place of another consonant.

Now five consonants T, R, N, G, L can be arranged among themselves in $\underline{5}$ ways whereas three vowels I, A, E can be arranged among themselves in $\underline{3}$ ways.

$$\begin{aligned}\therefore \text{required number of words} &= \underline{5} \times \underline{3} \\ &= (5 \times 4 \times 3 \times 2 \times 1) \times (3 \times 2 \times 1) \\ &= 120 \times 6 = 720.\end{aligned}$$

Example 9. How many different 4-digit numbers can be formed from the digits 2, 3, 4, 6 if each digit is used only once in a number? Further, how many of these numbers

- (i) end in a 4? (ii) end in a 3? (iii) end in a 3 or 6?

Sol. Given digits are 2, 3, 4, 6

$$\therefore \text{total number of digits} = 4$$

Number of digits to be taken at a time = 4

$$\therefore \text{required numbers} = {}^4P_4 = 4 \times 3 \times 2 \times 1 = 24$$

(i) Numbers end with 4

$\therefore$ fix 4 in the end

$\therefore$ 3 digits are to be arranged in 3 places

$$\therefore \text{required numbers} = {}^3P_3 = 3 \times 2 \times 1 = 6$$

(ii) Since numbers end with 3

$\therefore$ fix 3 in the end.

$\therefore$ 3 digits are to be arranged in 3 places

$$\therefore \text{required numbers} = {}^3P_3 = 3 \times 2 \times 1 = 6$$

(iii) Numbers having 3 in the end = 6

Similarly numbers having 6 in the end = 6

$$\therefore \text{required numbers} = 6 + 6 = 12$$

Example 10. How many different numbers between 100 and 1000 can be formed from the digits 0, 1, 2, 3, 4, 5 and 6, assuming that in a number, the digits cannot be repeated? How many of these will be divisible by 5?

Sol. Numbers between 100 and 1000 consist of three digits.

Given digits are 0, 1, 2, 3, 4, 5, 6.

$\therefore$ number of given digits = 7

Number of digits to be taken at a time = 3

$\therefore$ numbers formed of three digits = ${}^7P_3 = 7 \times 6 \times 5 = 210$

But these include those numbers which have '0' on their extreme left and these will be numbers of 2 digits and not of three digits.

$\therefore$ we have to exclude these from the total.

In order to find numbers which have '0' on the extreme left position, we fix '0' in that position and fill up the remaining two places out of six digits at our disposal, which can be done in 6P_2 ways i.e., 6×5 i.e., 30 ways.

$\therefore$ required number = $210 - 30 = 180$

Now numbers divisible by 5 must have either 5 or 0 in the unit's place.

In order to find number having 5 in unit's place we fix 5 in that place and fill up remaining 2 places with six remaining digits at our disposal, which can be done so in 6P_2 ways.

$\therefore$ numbers having 5 in the unit's place = ${}^6P_2 = 6 \times 5 = 30$

But these numbers also include those numbers which have '0' in the beginning i.e., numbers of two digits. So, in order to find numbers of 3 digits having 5 in the unit's place, we should subtract the numbers having 5 in the unit's place and '0' in the hundred's place and these numbers are ${}^5P_1 = 5$.

$\therefore$ numbers of 3 digits having 5 in the unit's place = $30 - 5 = 25$.

Now numbers of 3 digits having 0 in the unit place = 30.

$\therefore$ numbers divisible by 5 are $25 + 30 = 55$.

Example 11. How many numbers greater than 40000 can be formed using the digits 1, 2, 3, 4 and 5 if each digit is used only once in each number?

Sol. Given digits are 1, 2, 3, 4, 5

$\therefore$ number of given digits = 5

Number of digits to be taken at a time = 5

Since number is to be greater than 4000

$\therefore$ first digit from left should be either 4 or 5

$\therefore$ first place can be filled in 2 ways.

Remaining 4 places with 4 digits can be filled in 4P_4 ways

$\therefore$ required numbers = $2 \times {}^4P_4 = 2 \times (4 \times 3 \times 2 \times 1) = 2 \times 24 = 48$

Example 12. How many different signals can be formed with five given flags of different colours?

Sol. Number of flags = 5

A signal may be formed by hoisting any number of flags at a time.

Number of signals by hoisting one flag at a time = 5P_1

Number of signals by hoisting two flags at a time = 5P_2

Number of signals by hoisting three flags at a time = 5P_3

Number of signals by hoisting four flags at a time = 5P_4

Number of signals by hoisting five flags at a time = 5P_5

$\therefore$ total number of signals formed = ${}^5P_1 + {}^5P_2 + {}^5P_3 + {}^5P_4 + {}^5P_5$

$$= 5 + 5 \times 4 + 5 \times 4 \times 3 + 5 \times 4 \times 3 \times 2 + 5 \times 4 \times 3 \times 2 \times 1$$

$$= 5 + 20 + 60 + 120 + 120 = 325.$$

Example 13. 5 students are to be seated on 12 chairs in a line. In how many ways can this be done? (P.T.U. B.C.A.)

Sol. Total number of chairs = 12

Students to be seated = 5

Number of ways for seating = $P(12, 5)$

$$= \frac{12!}{7!} = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7!}{7!} = 95040$$

Example 14. How many ways can a student do a 10 questions true/false examination if he or she can choose not to answer any number of questions?

(P.T.U. B.C.A.)

Sol. Number of ways for attempting one question = 3 (true or false or leave)

Number of ways for attempting 10 questions = $3^{10} = 59049$

Example 15. In how many ways can the letters in word COMPUTER be arranged if the letters CO remain next to each other (in order) as a unit?

(Pbi.U., B.C.A.)

Sol. Number of letters in COMPUTER = 8

Since CO have to be together

Take CO as single letter

$\therefore$ number of letters = 7

$\therefore$ number of ways for arranging letters = $7!$

$$= 5040$$

[We do not multiply by 2 as we cannot write CO as OC]

EXERCISE 3 (d)

1.
 - (i) Ten horses are running a race. In how many ways can these horses come in the first, second and third place, assuming no ties ?
 - (ii) Seven songs are to be rendered in a programme. In how many different orders could they be rendered ?
 - (iii) There are six candidates contesting for a certain office in a municipal election. In how many ways can their names be listed on a ballot ?
 - (iv) Four books, one each in Chemistry, Physics, Biology and Mathematics are to be arranged in a shelf. In how many ways can this be done ?
 - (v) How many different signals can be generated from 6 flags of different colours if each signal makes use of all the flags at a time, placed one below the other ?
2. How many words, with or without meaning, can be formed using all the letters of the word EQUATION, using each letter exactly once ?
3. There are 6 items in column A and 6 items in column B. A student is asked to match each item in column A with an item in column B. How many possible answers (correct or incorrect) are there to the question ?
4.
 - (a) From a committee of 8 persons, in how many ways can we choose a chairman and vice chairman assuming one person cannot hold more than one position ?
 - (b) From a pool of 12 candidates, in how many ways can we select president, vice president, secretary and a treasurer if each of the 12 candidates can hold any office ?
5. How many different 5-letters words can be formed out of the letters of the word 'DELHI' ? How many of these will begin with D and end with E ?
6. The letters of the word TUESDAY are arranged in a line, each arrangement ending with letter S. How many different arrangements are possible ? How many of them start with letter D ?
7. Find the number of different 8-letters words formed from the letters of the word TRIANGLE if each word to have T and E at the end places.
8. Find the number of different 8- letters words formed from the letters of the word EQUATION, if each word is to start with a vowel.
9. In how many ways can 6 boys and 5 girls be arranged for a group photograph if the girls are to sit on chairs in a row and the boys are to stand, in row, behind them ?
10. Find the number of different 8-letter words formed from the letters of the word TRIANGLE if each word is to have vowels occupying the second, third and fourth places.

11. In how many ways can 4 boys and 3 girls be seated in a row of 7 chairs and girls alternate?
12. If there are six periods in each working day of a school, in how many ways can one arrange 5 subjects such that each subject is allowed at least one period?
13. Find the number of different 8-letter words formed from the letters of the TRIANGLE if each word is to have vowels occupying odd places.
14. In how many ways can the letters of the word 'PENCIL' be arranged so that E is always next to I?
15. Find the number of different 8-letter words formed from the letters of the TRIANGLE if each word is to have consonants never together.
16. In how many ways can 5 books on Chemistry and 4 books on Physics be arranged on a shelf so that the books on same subject remain together?
17. In how many ways can 5 children be arranged in a line such that (i) two of them, Ram and Shyam, are always together, (ii) two of them, Ram and Shyam, never together.
18. There are 8 students appearing in an examination, of which 3 have to appear in mathematics paper and the remaining 5 in different subjects. In how many ways can they be made to sit in a row if the candidates in mathematics cannot sit next to each other?
19. In how many ways can 4 boys and 3 girls be seated in a row so that two girls never sit together?
20. The principal wants to arrange 5 students on the platform such that the boy SALIM occupies the second position and such that the girls SITA is adjacent to the girl RITA. How many such arrangements are possible?
21. Three married couples are to be seated in a row having six seats in a cinema hall. If spouses are to be seated next to each other, in how many ways can they be seated? Find also the number of ways of their seating if all the ladies sit together.
22. A codeword is to consist of two distinct English alphabets followed by two distinct numbers between 1 and 9. For example CA 23 is a codeword. How many such codewords are there? How many of them end with an even number?
23. How many 4-digit numbers can be formed by using the digits 1 to 9 if repetition of digits is not allowed?
24. Find the number of 4-digit numbers that can be formed using the digits 1, 2, 3, 4, 5 if no digit is used more than once in a number. How many of these numbers will be even?
25. How many numbers of six digits can be formed from the digits 0, 1, 3, 5, 7, 9 when no digit is repeated? How many of them are divisible by 10?
26. How many four digit numbers are there with distinct digits?
27. How many three-digit numbers are there, with no digit repeated?

28. How many even numbers of three digits each can be made with the digits 1, 2, 3, 4, 6, 7 if no digit is repeated ?
29. How many odd numbers greater than 80000 can be formed using the digits 2, 3, 4, 5 and 8 if each digit is used only once in a number ?
30. How many even numbers are there with three digits such that if 5 is one of the digits, then 7 is the next digit ?
31. How many of the natural numbers from 1 to 1000 have none of their digits repeated ?

ANSWERS

- | | | | | |
|--------------|---------------|-------------------|-----------|---------|
| 1. (i) 720 | (ii) 5040 | (iii) 720 | (iv) 24 | (v) 720 |
| 2. 40320 | 3. 720 | 4. (a) 56 | (b) 11880 | |
| 5. 120 ; 6 | 6. 720 ; 120 | 7. 1440 | 8. 25200 | |
| 9. 86400 | 10. 720 | 11. 144 | 12. 1800 | |
| 13. 2880 | 14. 120 | 15. 37440 | 16. 5760 | |
| 17. (i) 48 | (ii) 72 | 18. 14400 | 19. 1440 | |
| 20. 8 | 21. 48 ; 144 | 22. 46800 ; 20800 | 23. 3024 | |
| 24. 120 ; 48 | 25. 600 ; 120 | 26. 4536 | 27. 648 | |
| 28. 60 | 29. 12 | 30. 365 | 31. 738 | |

Art-7. Find the number of permutations of n things taken all at a time when p of them are alike and of one kind, q of them are alike and of a second kind, all others being different.

Proof : Let n things be denoted by n letters, p of them being alike and denoted by 'a', q of them being alike and denoted by 'b' and the remaining being all different and denoted by $c, d, e, f, \dots$

Let x be the required number of permutations. Take any one of these permutations and replace p alike letters 'a' by ' p ' different letters $a_1, a_2, \dots, a_p$. These new p letters can be arranged among themselves in $\underline{p}$ new arrangements, when a 's are considered different. And, therefore, if such a change is made in all the x permutations, then the total number of permutations will be $x \times \underline{p}$.

Now, consider one of these $x \times \underline{p}$ arrangements, and replace q like letters by q different letters, $b_1, b_2, \dots, b_q$. These q letters can be arranged among themselves in $\underline{q}$ ways.

$\therefore$ one such permutation will give rise to $\underline{q}$ permutation and if such a change is made to all the $x \times \underline{p}$ permutations then the total number of permutations will be $x \times \underline{p} \times \underline{q}$

But number of permutations of n different things taken all at a time is $= \underline{n}$

$$\therefore \text{ we have } x \times \underline{p} \times \underline{q} = \underline{n} \Rightarrow x = \frac{\underline{n}}{\underline{p} \times \underline{q}}$$

Extension. This rule can be extended if in addition to the above r things are all of a third kind and so on.

ILLUSTRATIVE EXAMPLES

Example 1. In how many ways can 4 red, 3 yellow and 2 green discs be arranged in a row if the discs of the same colour are indistinguishable?

Sol.

Number of total discs	$= 4 + 3 + 2 = 9$
Number of red discs	$= 4$
Number of yellow discs	$= 3$
Number of green discs	$= 2$

$$\therefore \text{ required number of arrangements} = \frac{\underline{9}}{\underline{4} \cdot \underline{3} \cdot \underline{2}}$$

$$= \frac{9 \times 8 \times 7 \times 6 \times 5 \times \underline{4}}{\underline{4} \cdot (3 \times 2 \times 1) \times (2 \times 1)} = 1260$$

Example 2. In how many distinct ways can the product $x y^2 z^2$ be written without using exponents?

Sol. $x y^2 z^2$

Number of given letters $= 1 + 2 + 2 = 5$

Number of y 's $= 2$

Number of z 's $= 2$

$$\therefore \text{ required number of ways} = \frac{\underline{5}}{\underline{2} \times \underline{2}}$$

$$= \frac{5 \times 4 \times 3 \times 2 \times 1}{(2 \times 1) \times (2 \times 1)} = 30$$

Example 3. In how many distinct permutations of the letters in MISSISSIPPI do the first two I's not come together?

Sol. MISSISSIPPI

Number of given letters $= 11$

Number of I's $= 4$

Number of S's $= 4$

Number of P's $= 2$

$$\therefore \text{ number of permutations} = \frac{11!}{4! \times 4! \times 2!} = \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4}{4 \times (4 \times 3 \times 2 \times 1) \times (2 \times 1)} = 34650$$

Consider the four I's as one letter

$$\therefore 8 \text{ letters can be arranged in } \frac{8!}{4! \times 2!}$$

Also four I's can be arranged among themselves in $\frac{4!}{4!}$ ways

$$\therefore \text{ number of permutations in which four I's are always together} = \frac{8!}{4! \times 2!} \times \frac{4!}{4!} = \frac{8 \times 7 \times 6 \times 5 \times 4}{4 \times (2 \times 1)} = 840$$

$$\therefore \text{ required number of permutations} = 34650 - 840 = 33810$$

Example 4. Find the number of words which can be made using all the letters of the word AGAIN. If these words are written as in a dictionary, what will be the fiftieth word?

Sol. AGAIN

Number of given letters = 5

Number of A's = 2

$$\therefore \text{ number of words formed} = \frac{5!}{2!} = \frac{5 \times 4 \times 3 \times 2}{2} = 60$$

$$\text{Number of words starting with A} = \frac{4!}{1!} = 4 \times 3 \times 2 \times 1 = 24$$

$$\text{Number of words starting with G} = \frac{4!}{2!} = \frac{4 \times 3 \times 2 \times 1}{2 \times 1} = 12$$

$$\text{Number of words starting with I} = \frac{4!}{2!} = 12$$

$$\text{Number of words formed so far} = 24 + 12 + 12 = 48$$

$\therefore$ 49th word is NAAGI

$\therefore$ 50th word is NAAIG.

Example 5. How many distinguishable permutations of letters in the word BANANA are there? (Pbi. U. BCA 2009)

Sol. Total number of letters in BANANA = 6

Repeated letters are

A = 3 times

N = 2 times

$$\text{Number of arrangements} = \frac{6!}{3! \times 2!} = 60$$

EXERCISE 3 (e)

1. There are 3 white, 4 red and 1 blue marbles in bag. They are drawn one by one and arranged in a row. Assuming that all the 8 marbles are drawn, determine the number of different arrangements if marbles of same colour are indistinguishable.
2. In how many ways can 5 flags, in which 3 are red, one is white and one is blue be arranged on a staff, one below the other, if flags of one colour are indistinguishable?
3. How many permutations of the letter of word APPLE are there?
4. In how many different ways, the letters of the word ALGEBRA can be arranged in a row if
 - (i) the two A's are together?
 - (ii) the two A's are not together?
5. Find how many arrangements can be made with the letters of the word 'MATHEMATICS'? In how many of them
 - (i) consonants occur together
 - (ii) vowels do not occur together?
6. If the different permutations of the word 'EXAMINATION' are listed as in a dictionary, how many items are there in this list before the first word starting with E?
7. How many 5-digit even numbers can be formed using the digits 1, 2, 5, 5, 4?

ANSWERS

- | | | | |
|------------------------------------|-----------|-------|-------------------|
| 1. 280 | 2. 20 | 3. 60 | 4. (i) 720 ; 1800 |
| 5. 4989600; (i) 75600 (ii) 4868640 | 6. 907200 | 7. 24 | |

Art-8. Prove from first principles that ${}^nP_r = r \times {}^{n-1}P_{r-1} + {}^{n-1}P_r$

Proof : The total number of permutations of n things taken r at a time can be divided into two groups :

- (i) Those which always include a particular thing,
- (ii) Those which do not contain that particular thing.

To find the number of permutations which contain a particular thing. We are to fill up r places with n person under the condition that one particular person is to be placed definitely. That particular person can be seated in r different ways as he can be seated in any place. After he has been seated, remaining $r-1$ places can be filled up in ${}^{n-1}P_{r-1}$ ways.

$\therefore$ number of permutations containing a particular thing $= r \times {}^{n-1}P_{r-1}$

To find the number of permutations which do not contain that particular thing.

$\therefore$ a particular thing is to be excluded

$\therefore$ we are to fill up r places with $(n-1)$ persons.

$\therefore$ number of arrangements which do not contain a particular thing $= {}^{n-1}P_r$.

$\therefore$ we have, ${}^nP_r = r \times {}^{n-1}P_{r-1} + {}^{n-1}P_r$.

Art-9. Find the number of permutations of n different things taken r at a time when anything may be repeated any number of times in any arrangement.

Proof : We know that the number of permutations of n things taken r at a time is the same as the number of ways in which r places can be filled up with n things.

The first place can be filled up in n ways as any of the n things can be placed there. After filling the first place in any one of n ways, the second place can also be filled up in n ways as things can be repeated. Thus the two places taken together can be filled up in $n \times n$ i.e., n^2 ways. After filling up the first two places in any one of n^2 ways, the third place can also be filled up in n ways and consequently the three places taken together can be filled up in $n^2 \times n$, i.e., n^3 ways.

Proceeding in this way, we observe that the index of n is the same as the number of places filled up at this stage. Therefore, the number of ways filling up r places $= n^r$.

$\therefore$ the required number of permutations $= n^r$.

ILLUSTRATIVE EXAMPLES

Example 1. In how many ways four prizes can be distributed amongst five girls when there is no restriction as to the number of prizes any girl gets ?

Sol. First prize can be distributed in 5 ways as it can be awarded to any girl. Similarly each of the remaining three prizes can be awarded in 5 different ways.

$\therefore$ required number of ways $= 5 \times 5 \times 5 \times 5 = 5^4 = 625$.

Example 2. Find the number of permutation of n things taken r at a time in which 2 given things always occur.

In how many permutations they are always excluded ?

Sol. Number of permutations of n things taken r at a time is same as the number of ways of filling up r places with n things.

$\therefore$ 2 given things always occur

$\therefore$ first of all we arrange these two things which can be arranged in r places in rP_2 ways.

Also remaining $r-2$ places can be filled up with remaining $n-2$ things in ${}^{n-2}P_{r-2}$ ways.

$\therefore$ number of permutations in which two things are always included

$$= {}^rP_2 \times {}^{n-2}P_{r-2} = \frac{|r|}{|r-2|} \times \frac{|n-2|}{|(n-2)-(r-2)|} = \frac{|r|}{|r-2|} \times \frac{|n-2|}{|n-r|}$$

if two things are not to be included at all, then we are to fill up r places with $(n-2)$ things.

$$\therefore \text{ required number of permutations } = {}^{n-2}P_r = \frac{|n-2|}{|n-r-2|}$$

EXERCISE 3 (f)

1. In how many ways can 6 apples be distributed among three boys, when there is no restriction to the number of apples each boy receives?
2. How many natural numbers not exceeding 4321 can be formed with the digits 1, 2, 3 and 4, if the digits can repeat?
3. Ten different letters of an alphabet are given. Words with 5 letters are formed from these given letters. Prove that the number of words which have at least one letter repeated is 69760.

ANSWERS

1. 729

2. 313

Art-10. Circular Permutations

Find the number of ways in which n persons can be arranged at a round table.

Proof : When n persons are sitting around a circular table, then there is no first and last person. Let us fix the position of one person. The remaining $(n-1)$ persons can now be arranged in the remaining $(n-1)$ places in ${}^{n-1}P_{n-1}$ i.e., $(n-1)!$ ways.

$\therefore$ required number of ways = $(n-1)!$

Art-11. Clockwise and Anti-clockwise permutations

The total number of circular permutations can be divided into two types :

(i) Clockwise (ii) Anti-clockwise.

In two such arrangements each person has the same neighbour though in the reverse order and either of these arrangements can be obtained from the other by just over-turning the circle. If in this case, no distinction is made between clockwise and anti-clockwise arrangements then the two such arrangements are considered as only one distinct arrangement.

Hence the number of circular permutations in such cases = $\frac{1}{2} (n-1)!$

Note : Questions on necklaces with beads of different colours are to be tackled by the above formula, as in this case also there is no difference between clockwise and anti-clockwise arrangements.

ILLUSTRATIVE EXAMPLES

Example 1. In how many ways can 6 beads of different colours form a necklace?

Sol. Number of beads = 6

$$\begin{aligned}\therefore \text{required number of necklaces} &= \frac{1}{2} \underline{6-1} = \frac{1}{2} \cdot \underline{5} \\ &= \frac{1}{2} \times (5 \times 4 \times 3 \times 2 \times 1) = 60.\end{aligned}$$

Example 2. In how many ways can 5 persons A, B, C, D and E sit around a circular table if

- (i) B and D sit next to each other ?
 (ii) A and D do not sit next to each other ?

Sol. Given person are A, B, C, D, E

$$\therefore \text{number of given person} = 5$$

$$\therefore \text{number of ways in which they can be arranged in a circular table}$$

$$= \underline{5-1} = \underline{4} = 4 \times 3 \times 2 \times 1 = 24$$

(i) Consider two persons B and D as one. Therefore 4 persons can be arranged in circle in $\underline{4-1} = \underline{3}$ ways. Also two persons can be arranged among themselves, in $\underline{2}$ ways.

$$\therefore \text{number of ways in which two persons are always together}$$

$$= \underline{3} \times \underline{2}$$

$$= (3 \times 2 \times 1) \times (2 \times 1) = 6 \times 2 = 12$$

(ii) Consider two persons A and D as one. Therefore 4 persons can be arranged in circle in $\underline{4-1} = \underline{3}$ ways. Also two persons can be arranged among themselves, in $\underline{2}$ ways.

$$\therefore \text{number of ways in which two persons are always together}$$

$$= \underline{3} \times \underline{2}$$

$$= (3 \times 2 \times 1) \times (2 \times 1) = 6 \times 2 = 12$$

$$\therefore \text{required number of ways} = 24 - 12 = 12.$$

Example 3. Five girls and five boys are to be seated on a bench with the boys and girls alternating. Find the number of ways of their seating. In how many different ways could they sit around a circular table so that the boys and girls alternate ?

Sol. Total number of boys = 5

Total number of girls = 5

$\therefore$ boys and girls are to be seated alternatingly.

If boy takes first seat, then 5 boys can be arranged in 5 seats in $\underline{5}$ ways and 5 girls can be arranged in 5 seats in $\underline{5}$ ways

$$\therefore \text{number of arrangements} = \underline{5} \times \underline{5}$$

Similarly if girl takes first seat, then number of arrangements

$$= \underline{5} \times \underline{5}$$

$$\therefore \text{total number of arrangements} = \underline{5} \times \underline{5} + \underline{5} \times \underline{5}$$

$$= 120 \times 120 \times + 120 \times 120 \quad [\because \underline{5} = 5 \times 4 \times 3 \times 2 \times 1]$$

$$= 14400 + 14400 = 28800$$

Circular : Let the boys (or girls) be seated firstly leaving one seat vacant in between of two boys for a girl (or boy).

$$\text{This can be done in } \underline{5-1} = \underline{4} \text{ ways}$$

Now 5 girls (or boys) can be arranged in 5 vacant seats 5P_5 i.e. $\underline{5}$ ways

$$\therefore \text{required number of ways} = \underline{4} \times \underline{5}$$

$$= (4 \times 3 \times 2 \times 1) \times (5 \times 4 \times 3 \times 2 \times 1)$$

$$= 24 \times 120 = 2880$$

EXERCISE 3 (g)

- Four persons A, B, C and D are to be seated at a circular table. In how many ways can they be seated?
- Three boys and three girls are to be seated around a table in a circle. Among them, the boy A does not want any girl neighbour and the girl B does not want any boy neighbour. How many such arrangements are possible?
- In how many ways can 8 girls be seated at a round table provided Parveen and Vipul are not to sit together?
- In how many ways can 5 boys and 5 girls be seated at a round table, so that no two girls sit together?
- In how many ways 4 boys and 4 girls be seated at a round table provided that a boy is to be between two girls?
- There are 5 gentleman and 4 ladies at a round table. In how many ways can they seat themselves so that no two ladies are together?
- The Principals of six colleges seat themselves round a table to discuss student's unrest problem. In how many arrangements, Principals of X-college and Y-college will not sit together?

ANSWERS

1. 6

2. 4

3. 3600

4. 2880

5. 144

6. 2880

7. 72

• Multiplying above equations in (A) we get,

$${}^n P_r = (n-r+1)(n-r+2)\dots(n-1) \times {}^n P_1$$

$$= (n-r+1)(n-r+2)\dots(n-1) \cdot n$$

$${}^n P_r = n(n-1)\dots(n-r+1)$$

Art-14. Combination

In combination, we are not interested in arranging, but only in selecting r from n objects. In fact, we do not want to specify the ordering of these selected objects.

Differences between a permutation and a combination :

1. In a combination only a selection is made whereas in a permutation no selection is made, but also an arrangement in a definite order.

2. In a combination, the ordering of the selected objects is immaterial whereas in permutation, this ordering is essential.

Art-15. Find the number of combinations of n dissimilar things taken r at a time (i.e. the value of ${}^n P_r$).

Proof : Let the required number of combinations of n things taken r at a time be x .

$\therefore$ each combination contains r things, which can be arranged among themselves in $r!$ ways.

$\therefore$ corresponding to each combination, we get $r!$ permutations.

$\therefore$ total number of permutations due to x combinations $= x \cdot r!$

But number of permutations of n things taken r at a time $= {}^n P_r$.

$$\therefore \text{ we have, } x \cdot r! = {}^n P_r \quad \text{or} \quad x = \frac{{}^n P_r}{r!}$$

$$\text{i.e.,} \quad {}^n C_r = \frac{{}^n P_r}{r!} \Rightarrow {}^n C_r = \frac{n!}{r! (n-r)!}$$

$$\text{Cor. } {}^n C_r = \frac{{}^n P_r}{r!} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}$$

Art-16. Find the number of combinations of n things taken r at a time

(without using value)

OR

Prove from first principles that ${}^n C_r = \frac{n}{r} \cdot {}^{n-1} C_{r-1}$, and hence deduce the value of ${}^n C_r$.

Proof : Let the n things be denoted by n letters $a_1, a_2, \dots, a_n$.

$$= \frac{\underline{n}}{\underline{r-1} \underline{n-r}} \left[\frac{n+1}{r(n-r+1)} \right]$$

$$= \frac{(n+1) \underline{n}}{(r \underline{r-1}) (n-r+1) \underline{n-r}} = \frac{\underline{n+1}}{\underline{r} \underline{n-r+1}}$$

$\therefore$ L.H.S. = R.H.S.

DIVISION INTO GROUPS

Art-19. To find the number of ways in which $(m+n)$ things can be divided into two groups containing m and n things respectively.

Proof. When m things are selected out of $(m+n)$ things, then n things are automatically left behind and we get the required divisions into two groups.

The required number of ways division of groups = ${}^{m+n}C_m = \frac{\underline{m+n}}{\underline{m} \underline{n}}$

Cor. 1. If we have to divide $2m$ things into two equal distinct groups, then we have the required numbers of ways

$$= {}^{2m}C_m = \frac{\underline{2m}}{(\underline{m})^2}$$

Cor. 2. If no distinction is made between the groups, the groups will be interchanged in $\underline{2}$ ways without giving a new division.

$\therefore$ required number of ways = $\frac{\underline{2m}}{\underline{2} (\underline{m})^2}$

Art-20. Find the number of ways in which $(m+n+p)$ things can be divided into three groups containing m , n and p things respectively.

Proof. First of all, we divide $(m+n+p)$ things into two groups containing $(m+n)$ and p things respectively.

This can be done in ${}^{m+n+p}C_{m+n} = \frac{\underline{m+n+p}}{\underline{m+n} \underline{p}}$

Further, we divide each group of $(m+n)$ things into two groups of m and n things respectively.

This can be done in ${}^{m+n}C_m = \frac{\underline{m+n}}{\underline{m} \underline{n}}$ ways

By the Principle of Association, the required number of ways in which $(m+n+p)$ things can be divided into three groups of m , n and p things respectively

$$= \frac{\underline{m+n+p}}{\underline{m+n} \underline{p}} \times \frac{\underline{m+n}}{\underline{m} \underline{n}} = \frac{\underline{m+n+p}}{\underline{m} \underline{n} \underline{p}}$$

ILLUSTRATIVE EXAMPLES

Example 1. Verify that $C(8, 4) + C(8, 3) = C(9, 4)$.

Sol. L.H.S. = $C(8, 4) + C(8, 3) = {}^8C_4 + {}^8C_3$

$$= \frac{8 \times 7 \times 6 \times 5}{1 \times 2 \times 3 \times 4} + \frac{8 \times 7 \times 6}{1 \times 2 \times 3} = 70 + 56 = 126$$

$$\text{R.H.S.} = C(9, 4) = {}^9C_4 = \frac{9 \times 8 \times 7 \times 6}{1 \times 2 \times 3 \times 4} = 126$$

$\therefore$ L.H.S. = R.H.S.

Example 2. Evaluate ${}^{50}C_{47}$, $C(15, 14)$.

Sol. Here ${}^{50}C_{47} = {}^{50}C_3$ $[\because {}^nC_r = {}^nC_{n-r}]$

$$= \frac{50 \times 49 \times 48}{1 \times 2 \times 3} = 19600$$

Again $C(15, 14) = {}^{15}C_{14} = {}^{15}C_1$ $[\because {}^nC_r = {}^nC_{n-r}]$

$$= \frac{15}{1} = 15$$

Example 3. If $C(n, 10) = C(n, 12)$, determine n and hence $C(n, 5)$.

Sol. Since $C(n, 10) = C(n, 12)$

$$\therefore {}^nC_{10} = {}^nC_{12}$$

$$\Rightarrow \text{either } 10 = 12$$

which is impossible

$$\text{or } n = 10 + 12$$

$$\therefore n = 22$$

Now $C(n, 5) = C(22, 5) = {}^{22}C_5$

$$= \frac{22 \times 21 \times 20 \times 19 \times 18}{1 \times 2 \times 3 \times 4 \times 5} = 26334$$

Example 4. Prove that

$$C(2n, n) = \frac{2^n [1.3.5 \dots (2n-1)]}{\lfloor n \rfloor}$$

Sol.

$$\text{L.H.S.} = C(2n, n) = {}^{2n}C_n = \frac{\lfloor 2n \rfloor}{\lfloor n \rfloor \cdot \lfloor 2n - n \rfloor} = \frac{\lfloor 2n \rfloor}{\lfloor n \rfloor \cdot \lfloor n \rfloor}$$

$$= \frac{1.2.3.4.5 \dots (2n-1)(2n)}{(\lfloor n \rfloor)^2}$$

$$= \frac{[1.3.5 \dots (2n-1)][2.4.6 \dots (2n)]}{(\lfloor n \rfloor)^2}$$

$$\text{and } \frac{{}^nC_r}{{}^{n+1}C_r} = \frac{9}{13}$$

$$\Rightarrow \frac{\frac{{}^n C_r}{{}^{n+1} C_r}}{\frac{{}^r C_r}{{}^{n-r+1} C_r}} = \frac{9}{13}$$

$$\therefore \frac{{}^n C_r}{{}^r C_r} \times \frac{{}^{n-r+1} C_r}{{}^{n-r+1} C_r} = \frac{9}{13} \Rightarrow \frac{{}^n C_r}{(n+1) {}^n C_r} \times \frac{(n-r+1) {}^{n-r} C_r}{{}^{n-r} C_r} = \frac{9}{13}$$

$$\therefore \frac{{}^n C_r}{n+1} \times \frac{n-r+1}{{}^{n-r} C_r} = \frac{9}{13}$$

$$\therefore \frac{n-r+1}{n+1} = \frac{9}{13} \quad \Rightarrow \quad 13n - 13r + 13 = 9n + 9$$

$$\therefore 4n - 13r + 4 = 0$$

$$\therefore 4(3r) - 13r + 4 = 0 \quad [\because \text{of (1)}]$$

$$\therefore 12r - 13r + 4 = 0$$

$$\Rightarrow -r + 4 = 0 \quad \Rightarrow r = 4$$

$$\therefore \text{from (1), } n = 3 \times 4 = 12$$

$$\therefore n = 12, r = 4$$

EXERCISE 3 (h)

1. Evaluate $C(19, 17) + C(19, 18)$.

2. (a) Evaluate each of the following :

$$(i) {}^{10}C_4 + {}^{10}C_5 \quad (ii) {}^{61}C_{57} - {}^{60}C_{56}$$

(b) Evaluate :

$$(i) {}^{13}C_6 + {}^{13}C_5 \quad (ii) {}^{19}C_{17} + {}^{19}C_{18}$$

$$(iii) {}^{25}C_{22} - {}^{24}C_{21} \quad (iv) {}^{31}C_{26} - {}^{30}C_{26}$$

(c) Prove that ${}^2C_1 + {}^3C_1 + {}^4C_1 = {}^3C_2 + {}^4C_2$

(d) Prove that $1 + {}^3C_1 + {}^4C_2 = {}^5C_3$.

3. Prove that

$$(i) 1 + C(3, 1) + C(4, 2) = C(5, 3)$$

$$(ii) C(2, 1) + C(3, 1) + C(4, 1) = C(3, 2) + C(4, 2)$$

4. Determine n if $C(2n, 3) : C(n, 2) = 12 : 1$.
5. Determine n if $C(2n, 3) : C(n, 3) = 11 : 1$.
6. If ${}^{15}C_3 r = {}^{15}C_{r+3}$, find r .
7. (i) If $C(n, 7) = C(n, 5)$, find n .
(ii) If $C(n, 8) = C(n, 6)$, find $C(n, 2)$.
8. Prove that
 - (i) $n C(n-1, r-1) = (n-r+1) C(n, r-1)$
 - (ii) $r C(n, r) = n C(n-1, r-1)$
 - (iii) $(n-r) C(n, r) = (r+1) C(n, r+1)$
 - (iv) ${}^nC_r \times {}^rC_s = {}^nC_s \times {}^{n-s}C_{r-s}$
9. If $m = C(n, 2)$, prove that $C(m, 2) = 3 C(n+1, 4)$.
10. Show that $\frac{(n+1)(n+2)(n+3)\dots(n+r)}{r!}$ is positive integer.
11. Prove that the product of k consecutive natural numbers is divisible by $k!$.
12. If ${}^nC_x = 56$ and ${}^nP_x = 336$, find n and x .
13. If ${}^nC_r : {}^nC_{r+1} = 1 : 2$ and ${}^nC_{r+1} : {}^nC_{r+2} = 2 : 3$, find n and r .

ANSWERS

1. 190
2. (a) (i) 462 (ii) 34220
(b) (i) 3003 (ii) 190 (iii) 276 (iv) 142506
4. 5 5. 6 6. 3
7. (i) 12 (ii) 14 12. $x = 3, n = 8$ 13. $n = 14, r = 4$

Art-23. Practical Problems Involving Combinations

ILLUSTRATIVE EXAMPLES

Example 1. In how many ways can a committee be selected from 15 persons if the committee is to have

- (i) 3 members.
- (ii) 13 members.

Sol. Total number of persons = 15

(i) Number of persons to be selected = 3

$$\therefore \text{required number of ways} = {}^{15}C_3 = \frac{15 \times 14 \times 13}{1 \times 2 \times 3} = 455$$

(ii) Number of persons to be selected = 13

$$\therefore \text{required number of ways} = {}^{15}C_3 = {}^{15}C_2 = \frac{15 \times 14}{1 \times 2} = 105$$

Example 2. In how many ways can a student choose a programme of 5 courses if 9 courses are available and 2 courses are compulsory for every student?

Sol. Total number of courses = 9

Number of courses to be chosen = 5

$\therefore$ two courses are compulsory

$\therefore$ student is to select 3 courses out of 7 courses

$$\therefore \text{required number of ways} = {}^7C_3 = \frac{7 \times 6 \times 5}{1 \times 2 \times 3} = 35$$

Example 3. Determine the number of 5-card combinations out of a deck of 52 cards if atleast one of the five cards has to be an king.

Sol. Total number of cards = 52

Number of king = 4

$\therefore$ remaining cards = 48

Number of combinations of 5 cards out of 52 = ${}^{52}C_5$

$$= \frac{52 \times 51 \times 50 \times 49 \times 48}{1 \times 2 \times 3 \times 4 \times 5}$$

Number of combinations of 5 cards containing no king

$$= {}^{48}C_5 = \frac{48 \times 47 \times 46 \times 45 \times 44}{1 \times 2 \times 3 \times 4 \times 5}$$

$\therefore$ required number of combinations

$$= \frac{52 \times 51 \times 50 \times 49 \times 48}{1 \times 2 \times 3 \times 4 \times 5} - \frac{48 \times 47 \times 46 \times 45 \times 44}{1 \times 2 \times 3 \times 4 \times 5}$$

$$= 886656$$

Example 4. A question paper has two parts, part A and part B, each containing 10 questions. If the student has to choose 8 questions from part A and 5 from part B, in how many ways can he choose the questions?

Sol. Number of questions in part A = 10

Number of questions in part B = 10

8 questions out of 10 of part A can be selected in ${}^{10}C_8$ ways. Also 5 questions out of 10 of part B can be selected in ${}^{10}C_5$ ways.

Part A	Part B
10	10
8	5

$$\begin{aligned}
 \therefore \text{required number of ways} &= {}^{10}C_8 \times {}^{10}C_5 \\
 &= {}^{10}C_2 \times {}^{10}C_5 = \frac{10 \times 9}{1 \times 2} \times \frac{10 \times 9 \times 8 \times 7 \times 6}{1 \times 2 \times 3 \times 4 \times 5} \\
 &= 45 \times 252 = 11340
 \end{aligned}$$

Example 5. In an examination, Yamini has to select 4 questions from each part. There are 6, 7 and 8 questions in Part I, Part II and Part III, respectively. What is the number of possible combinations in which she can choose the questions?

Sol. Number of questions in Part I = 6

Number of question in Part II = 7

Number of questions in Part III = 8

Now 4 question out of 6 of part I can be selected in 6C_4 ways, 4 questions out of 7 of part II can be selected in 7C_4 ways and 4 question out of 8 of part III can be selected in 8C_4 ways.

$$\therefore \text{required number of combinations} = {}^6C_4 \times {}^7C_4 \times {}^8C_4$$

$$= {}^6C_2 \times {}^7C_3 \times {}^8C_4$$

$$= \frac{6 \times 5}{1 \times 2} \times \frac{7 \times 6 \times 5}{1 \times 2 \times 3} \times \frac{8 \times 7 \times 6 \times 5}{1 \times 2 \times 3 \times 4} = 36750$$

Example 6. There are 15 points in a plane, no three of which are in the same straight line excepting 4, which are collinear. Find the number of (i) straight lines (ii) triangles formed by joining them.

Sol. (i) We know that join of any two distinct points, gives a line.

$$\therefore \text{number of lines got from 15 points, no three of which are collinear} \\ = {}^{15}C_2 = \frac{15 \times 14}{1 \times 2} = 105$$

$$\text{Lines got from 4 points} = {}^4C_2 = \frac{4 \times 3}{1 \times 2} = 6$$

$$\therefore \text{number of lines lost due to 4 collinear points} = 6 - 1 = 5$$

$$\therefore \text{required number of lines} = 105 - 5 = 100 \quad [\because 4 \text{ collinear points do give one line}]$$

(ii) We know that any three points give a triangle.

$$\therefore \text{number of triangles got from, 15 points no three of which are collinear}$$

$$= {}^{15}C_3 = \frac{15 \times 14 \times 13}{1 \times 2 \times 3} = 455$$

$$\text{Triangles got from 4 points} = {}^4C_3 = {}^4C_1 = 4$$

$$\therefore \text{number of triangles lost due to collinear points} = 4$$

$$\therefore \text{required number of triangles} = 455 - 4 = 451.$$

Example 7. In class XI examination, a candidate has to pass in all the five subjects taken in order to be declared as pass. In how many ways can he fail ?

Sol. Number of subjects = 5

Candidate will fail if he fail in one or two or three or four or five subjects.

$$\begin{aligned}\therefore \text{required number of ways} &= {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 \\ &= \frac{5}{1} + \frac{5 \times 4}{1 \times 2} + \frac{5 \times 4}{1 \times 2} + \frac{5}{1} + 1 \\ &= 5 + 10 + 10 + 5 + 1 = 31\end{aligned}$$

Example 8. A mathematics paper consists of 10 questions divided into two parts I and II. Each part containing 5 questions. A student is required to attempt 6 questions in all, taking at least 2 questions from each part. In how many ways can the student select the questions ?

Sol. Number of questions in part I = 5

Number of question in part II = 5

Number of questions to be attempted = 6

$\therefore$ each selection contains at least 2 from each part

$\therefore$ different possibilities are

(i) 2 from part I, 4 from part II

3 from part I, 3 from part II

4 from part I, 2 from part II

$$\begin{aligned}\therefore \text{required number of ways} &= {}^5C_2 \times {}^5C_4 + {}^5C_3 \times {}^5C_3 + {}^5C_4 \times {}^5C_2 \\ &= {}^5C_2 \times {}^5C_1 + {}^5C_2 \times {}^5C_2 + {}^5C_1 \times {}^5C_2 \\ &= \frac{5 \times 4}{1 \times 2} \times \frac{5}{1} + \frac{5 \times 4}{1 \times 2} \times \frac{5 \times 4}{1 \times 2} + \frac{5}{1} \times \frac{5 \times 4}{1 \times 2} \\ &= 10 \times 5 + 10 \times 10 + 5 \times 10 = 50 + 100 + 50 = 200\end{aligned}$$

Example 9. A group consists of 4 girls and 7 boys. In how many ways can a team of 5 members be selected if the team has

(i) no girls ?

(ii) at least one boy and one girl ?

(iii) at least three girls ?

Sol. Number of boys = 7

Number of girls = 4

Committee is to consist of 5

(i) $\therefore$ committee includes no girl

$\therefore$ we are to select 5 boys out of 7 boys

Part I	Part II
5	5
2	4
3	3
4	2

Boys	Girls
7	4
5	

$$\therefore \text{number of committees} = {}^7C_5 = {}^7C_2 = \frac{7 \times 6}{1 \times 2} = 21$$

(ii) $\therefore$ committee is to include at least one boy and one girl

$\therefore$ different possibilities are

(a) 1 boy, 4 girls

(b) 2 boys, 3 girls

(c) 3 boys, 2 girls

(d) 4 boys, 1 girl

Boys	Girls
7	4
1	4
2	3
3	2
4	1

$\therefore$ required number of ways

$$= {}^7C_1 \times {}^4C_4 + {}^7C_2 \times {}^4C_3 + {}^7C_3 \times {}^4C_2 + {}^7C_4 \times {}^4C_1$$

$$= {}^7C_1 \times 1 + {}^7C_2 \times {}^4C_1 + {}^7C_3 \times {}^4C_2 + {}^7C_4 \times {}^4C_1$$

$$= 7 \times 1 + \frac{7 \times 6}{1 \times 2} \times \frac{4}{1} + \frac{7 \times 6 \times 5}{1 \times 2 \times 3} \times \frac{4 \times 3}{1 \times 2} + \frac{7 \times 6 \times 5}{1 \times 2 \times 3} \times \frac{4}{1}$$

$$= 7 + 84 + 210 + 140 = 441$$

(iii) $\therefore$ committee is to include

at least three girls

$\therefore$ different possibilities are

(a) 2 boys, 3 girls

(b) 1 boy, 4 girls

Boys	Girls
7	4
2	3
1	4

$$\therefore \text{required number of ways} = {}^7C_2 \times {}^4C_3 + {}^7C_1 \times {}^4C_4$$

$$= {}^7C_2 \times {}^4C_1 + {}^7C_1 \times 1$$

$$= \frac{7 \times 6}{1 \times 2} \times \frac{4}{1} \times \frac{7}{1} \times 1 = 84 + 7 = 91$$

Example 10. How many words each of 3 vowels and 2 consonants, can be formed from the letters of the word 'INVOLUTE'?

Sol. Given word is INVOLUTE.

Here N, V, L, T are consonants and I, O, U, E are vowels.

$\therefore$ number of consonants = 4

and number of vowels = 4

2 consonants can be selected out of 4 consonants in 4C_2 ways and 3 vowels out of 4 vowels can be selected in 4C_3 ways.

$$\therefore \text{total number of groups formed} = {}^4C_2 \times {}^4C_3$$

$$= \frac{4 \times 3}{1 \times 2} \times \frac{4 \times 3 \times 2}{1 \times 2 \times 3} = 6 \times 4 = 24$$

Now each group contains 5 letters which can be arranged among themselves in 5 ways.

$$\therefore \text{required number of words formed} = 24 \times 5$$

$$= 24 \times (5 \times 4 \times 3 \times 2 \times 1)$$

$$= 24 \times 120 = 2880$$

Example 11. The judiciary committee at a college is made up of 3 faculty members and 4 students. If 10 faculty members and 25 students have been nominated for committee, how many judiciary committee are possible? (P.T.U. B.C.A. I 2005)

Sol. Number of ways for faculty members = $C(10, 3)$

Number of ways for students = $C(25, 4)$

Total ways = $C(10, 3) \times C(25, 4)$

$$= \frac{{}^{10}P_3}{{}^3P_3} \times \frac{{}^{25}P_4}{{}^4P_4} = \frac{10 \times 9 \times 8 \times 7}{7 \times 3 \times 2 \times 1} \times \frac{25 \times 24 \times 23 \times 22 \times 21}{21 \times 4 \times 3 \times 2 \times 1}$$

$$= 10 \times 3 \times 4 \times 25 \times 23 \times 22 = 1518000$$

Example 12. A delegation of 4 students is to be selected out of 12. In how many ways it can be done if there are two students who refuse to be in delegation together. (P.T.U. B.C.A. I 2007)

Sol. First we find total number of delegations

i.e. to select 4 students out of 12

Number of ways = $C(12, 4)$

$$= \frac{{}^{12}P_4}{{}^4P_4} = \frac{12 \times 11 \times 10 \times 9 \times 8}{8 \times 4 \times 3 \times 2 \times 1} = 495$$

Now we find ways in which two students are together. For this two students are taken from 2 and remaining two are taken from 10

Number of ways = $C(2, 2) \times C(10, 2)$

$$= 1 \times \frac{{}^{10}P_2}{{}^2P_2} = 45$$

Number of ways in which two students are not together = $495 - 45 = 450$

Example 13. How many five persons committee constituted from a group of six men and five women consisting of

- (i) at least one man (ii) at most one man?

(Pbi.U., B.C.A., 2008)

Example 15. In how many ways can 3 integers be selected from integers 1, 2, 3, ..., 30 so that their sum is even ?
(Pbi.U., B.C.A., 2006)

Sol. Total number of integers = 30

We have to select 3 integers so that their sum is even.

For this we have two cases :

I. Select all even numbers

II. Select 2 odd and one even number

Now in integers 1 to 30

Number of even integers = 15

Number of odd integers = 15

Number of ways for case I = ${}^{15}C_3$

Number of ways for case II = ${}^{15}C_2 \times {}^{15}C_1$

$$\begin{aligned} \text{Total ways} &= {}^{15}C_3 + {}^{15}C_2 \times {}^{15}C_1 = \frac{{}^{15}P_3}{3!} + \frac{{}^{15}P_2}{2!} \times 15 \\ &= 455 + 105 \times 15 = 455 + 1575 = 2030 \end{aligned}$$

EXERCISE 3 (i)

- A committee of 2 boys is to be selected from 4 boys. In how many ways can this be done ?
 - How many different teams of 7 players can be chosen from 10 players ?
 - How many selections of 4 books can be made from 8 different books ?
 - Sudha wants to choose any 9 stamps from a set of 11 different stamps. How many different selections can she make ?
 - Twelve persons meet in a room and each shakes hand with all the others. Find the number of hand-shakes.
- In a meeting after every one hand shaken hands with every one else, it was found that 66 hand shakes were exchanged. How many members were present at the meeting ?
- In an examination, a student is to answer 4 questions out of 5. Question 1 and 2 are, however, compulsory. Determine the number of ways in which the student can make the choice.
- In an examination paper on mathematics 10 questions are set. In how many different ways can you choose 6 questions to answer ? If, however, question number 1 is made compulsory, in how many ways can you select to answer 6 questions in all ?
- We wish to select 6 persons from 8, but if the person A is chosen, then B must be chosen. In how many ways can the selection be made ?

6. Determine the number of 5 cards combinations out of a deck of 52 cards is exactly one ace in each combination.
7. A father with eight children takes them 3 at a time to zoological garden as he can take without the same three children together more than once. How often will each child go? How often will the father go?
8. In how many ways can a cricket eleven be selected out of 15 players?
(a) In how many of them a particular player be excluded?
(b) In how many of them will he be included?
9. In how many ways can we select a cricket eleven from 17 players in which 5 players can bowl if each cricket eleven must include exactly 4 bowlers?
10. Show that the total number of ways in which six '+' and four '-' signs are arranged in a line such that no two '-' signs occur together is 35.
11. A bag contains 5 black and 6 red balls. Determine the number of ways in which 2 black and 3 red balls can be selected from lot.
12. A bookshelf contains 7 different mathematics textbooks and 5 different physics textbooks. How many groups of 3 mathematics and 3 physics textbooks can be selected?
13. There are 6 boys and 3 girls in a class. An entertainment committee of 5 is to be selected such that there are 4 boys and 2 girls in the committee. In how many ways can the committee be selected?
14. For the posts of 5 teachers, there are 23 applicants. 2 posts are reserved for SC candidates and there are 7 SC candidates among the applicants. In how many ways can the selection be made?
15. (a) In how many ways can a team of 3 boys and 3 girls be selected from 7 boys and 4 girls?
(b) How many different committees each consisting of 3 girls and 2 boys can be chosen from 7 girls and 5 boys?
16. Find the number of ways in which committee of 4 Americans, 3 Englishmen and 2 Frenchmen can be formed from a group consisting of 7 Americans, 6 Englishmen and 4 Frenchmen?
17. Find the number of ways of selecting 9 balls from 6 red balls, 5 white balls and 5 blue balls if each selection consists of 3 balls of each colour.
18. (a) How many lines can be drawn through 21 points on a circle?
(b) There are 15 points in a plane, no three of which are collinear. Find the number of triangles formed by joining them.
19. (i) Seven points lie on a circle. How many chords can be drawn between these points?
(ii) How many lines can be drawn through n points on a circle? How many triangles can be drawn through n points on a circle?
(iii) How many diagonals does a ten-sided polygon have? How many regions can be got by joining its vertices?
(iv) A number of points are marked on a plane and are connected pairwise by line segments. If the total number of line segments is 10, how many points are marked on the plane?

20.

21.

22.

23.

24.

25.

26.

27.

28.

29.

30.

31.

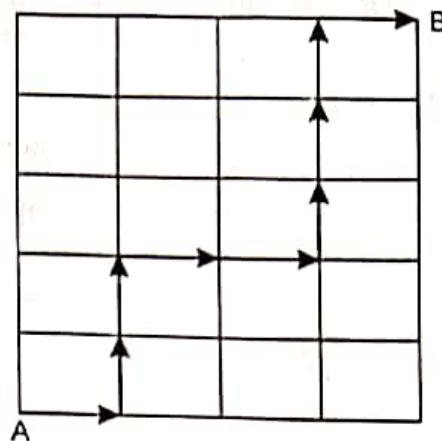
32.

33.

34.

20. The number of diagonals of a polygon is 20. Find the number of its sides.
21. In how many ways can 12 different balls be divided equally (i) into three heaps, (ii) among 3 boys.
22. Ram has 5 friends. In how many ways can he invite one or more of them to a party?
23. (i) A man has 6 friends. In how many ways can he invite one or more of them to a party?
(ii) In an examination, a candidate has to pass in each of four subjects taken by him. In how many ways can he fail?
24. A committee of 5 is to be selected from among 6 boys and 5 girls. Determine the number of ways of selecting the committee if it is to consist of at least 1 boy and 1 girl.
25. A committee of four has to be selected from among 6 boys and 5 girls. The committee is to include at least 1 boy and at least 1 girl. In how many ways can we select the committee?
26. A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consists of
(i) exactly 3 girls (ii) at least 3 girls
27. In an examination, a question paper consists of 12 questions divided into two parts I and II, containing 5 and 7 questions, respectively. A student is required to attempt 8 questions in all, selecting at least 3 from each part. In how many ways can a student select the questions?
28. A sports team of 11 students is to be constituted, choosing at least 5 from class XI and at least 5 from class XII. If there are 20 students in each of these classes, in how many ways can the team be constituted?
29. From a class of 12 boys and 8 girls, 8 students are to be chosen for a competition, at least including 4 boys and 4 girls. The 2 girls who won the prizes last year should be included. In how many ways can the selection be made?
30. In a village, there are 87 families of which 52 families have atmost 2 children. In a rural development programme, 20 families are to be choosen for assistance, of which at least 18 families must have atmost 2 children. In how many ways can the choice be made?
31. A boy has 3 library tickets and 8 books of his interest in the library. Of these 8, he does not want to borrow Chemistry Part II unless Chemistry Part I is also borrowed. In how many ways can he choose the three books to be borrowed?
32. From a class of 25 students, 10 are to be chosen for an excursion party. There are 3 students who decide that either all of them will join or none of them will join. In how many ways can they be chosen?
33. From 5 consonants and 4 vowels, how many words can be constructed using 3 consonants and 2 vowels?
34. The English alphabet has 5 vowels and 21 consonants. How many words with two different vowels and 2 different consonants can be framed from the alphabet?

4. A person is to walk from A to B. However, he is restricted to walk only to the right of A or upwards of A, but not necessarily in this order. One such path is shown in figure. Determine the total number of paths available to the person from A to B.



5. In how many ways can three jobs I, II and III be assigned to three persons A, B and C, if one person is assigned only one job and all are capable of doing each job? Which assignment of jobs will take the least time to complete the jobs, if time taken (in hours) by an individual on each job is as follows?

Job → Person ↓	I	II	III
A	5	4	4
B	$4\frac{1}{2}$	$3\frac{1}{2}$	4
C	5	3	5

ANSWERS

1. 999999 2. 720 3. $2^1, 2^2, 2^3, 2^4, 30$ (i) 32 (ii) 62
4. 126 5. 6; A → III, B → I, C → II

Art-25. The Pigeonhole Principle (Simple Form)

If n pigeons are assigned to m pigeonholes and $m < n$, then there is at least one pigeonhole that contains two or more pigeons.

Proof: Label n pigeons with the numbers 1 to n and m pigeonholes with the numbers 1 to m . Starting with pigeon 1 and pigeonhole 1, assign each pigeon in order to the pigeonhole with the same number. So we can assign as many pigeons as possible to distinct pigeonholes. Since the number m of pigeonholes is less than the number n of the pigeons, so $n - m$ pigeons are left that are not assigned to a pigeonhole. Therefore, there is atleast one pigeonhole that will be assigned one or more than one pigeon again.

∴ there is at least one pigeonhole that contains two or more pigeons.

Example 1. Show that if eight people are in the room, at least two of them have birthdays that occur on same day of the week. (Pbi.U., M.C.A.)

Sol. Total number of persons = 8

Total number of days in week = 7

Consider persons as pigeons and days as pigeonholes. As number of pigeons is more than pigeon holes so at least two pigeons will have same pigeonholes i.e. at least two persons will have birthday on same day of week.

Example 2. Use Pigeonhole Principle to show that if seven numbers from 1 to 13 are chosen, then two of them will add upto 13. (Pbi.U., B.A.)

Sol. The sets which add upto 13 are $\{1, 12\}$, $\{2, 11\}$, $\{3, 10\}$, $\{4, 9\}$, $\{5, 8\}$, $\{6, 7\}$.

By Pigeonhole principle, if we have to choose seven numbers then we must choose at least two numbers belonging to one set. Thus two of the seven numbers will add upto 13.

Example 3. Use Pigeonhole Principle to prove that an injection cannot exist from a finite set A and a finite set B if Cardinality of A is greater than Cardinality of B. (Pbi. U. M.C.A. 2001)

Sol. Let $n(A) = a$ $n(B) = b$ where $a > b$

Consider elements of Set B as pigeonholes and elements of set A as pigeons. As number of pigeons are more than pigeon holes so at least two pigeons will have same pigeonhole or we can say $\exists x, y \in A$ such that $f(x) = f(y)$. But $x \neq y$.

So $f: A \rightarrow B$ is not injective.

Example 4. Show that if any five numbers are chosen from 1 to 8, two of them will add upto 9.

Sol. The sets which add upto 9 are $\{1, 8\}$ $\{2, 7\}$ $\{3, 6\}$ $\{4, 5\}$. By Pigeonhole Principle, if we have to choose five numbers then we must take atleast two numbers belonging to one set..... thus two of five numbers will definitely add upto 9.

Example 5. Show that any 11 numbers chosen from the set $\{1, 2, 3, 4, \dots, 20\}$ then one will be multiple of other.

Sol. Each number can be represented by $(2^m) \times n$ where n is odd.

So now as there are only 10 odd numbers between 1 to 20, if we select 11 numbers from the set 1, 2, 3,, 18, 19, 20 then by Pigeon Hole Principle two of them will be bound to have same odd part.... Hence one of them can be divide by other. Hence Proved.

Example 6. If seven points are chosen in a regular hexagon each of side 1 unit, then one of them must be no farther apart than 1 unit.

Sol. We must find the pigeons and the pigeonholes... So divide the hexagon into four equal regions. Each will be an equilateral triangle of length of each side as one unit. These are our pigeonholes and the points as our pigeons. So by Pigeon Hole Principle, these seven points will be in the same triangle. Thus they will be at a distance less than 1 unit. Hence Prove.

Example 7. Shirts are numbered 1 to 20 are to be worn by the 20 members of a bowling league. When any three of these shirts are chosen, the league proposes to use the sum of their shirts as a code number for the team. Show that if any eight are chosen from the 20 shirts, then from these eight shirts one may form atleast two different teams having the same code number.

Sol. Number of ways of choosing a code for any selected 8 shirts is $C(8, 3) = 56$ different teams. Now these are our pigeons and the pigeonholes are the number of options we have. That is it will range from $(1 + 2 + 3) = 6$ to $(18 + 19 + 20) = 57$. Thus we have only 52 option. So by Pigeon Hole Principle, atleast two of them will have the same team codes.

Example 8. Show that if seven positive integers are chosen, two of them will have same remainder when divided by 6.

Sol. Seven positive integers divided by 6 will have seven remainders from 0 to 5. According to pigeonhole rule, two of these seven remainders will be same.

Art-26. Pigeonhole Principle (Extended Form)

If n pigeons are assigned m pigeonholes, where n is sufficiently large as compared to m , then one of the pigeonholes must contain at least $\left\lceil \frac{n-1}{m} \right\rceil + 1$ pigeons.

Proof : Assume that the result is false

$\therefore$ each pigeonhole does not contain more than $\left\lceil \frac{n-1}{m} \right\rceil$ pigeon.

$\therefore$ maximum possible number of pigeons = $\left\lceil \frac{n-1}{m} \right\rceil m \leq \frac{n-1}{m} \cdot m = n - 1$

This contradicts the given result that number of pigeons is n .

$\therefore$ our supposition is wrong.

Hence the result.

Example 9. Show that if 9 colours are used to paint 1000 houses, at least 112 houses will be of the same colour

Sol. Let n denote the number of given houses and m denote the number of colours.

Here $n = 1000, m = 9$

$\therefore \left\lceil \frac{n-1}{m} \right\rceil + 1 = \left\lceil \frac{1000-1}{9} \right\rceil + 1 = \left\lceil \frac{999}{9} \right\rceil + 1 = 111 + 1 = 112$

$\therefore$ at least 112 houses will be of the same colour.

Example 10. How many people among 200000 people are born at same time (hour, minute, seconds) ? Use Pigeonhole principle to find it. (Pbi. U. B.C.A. 2007)

Sol. Total number of persons = 200000

Total number of seconds in a day = $24 \times 60 \times 60 = 86,400$

Here, we have to assign a time to each person

So person are like pigeons time is like pigeonhole.

Number of pigeons (n) = 200000

Number of pigeonholes (m) = 86,400

$$\begin{aligned}
 \text{Min. number of persons having same birthday} &= \left\lfloor \frac{n-1}{m} \right\rfloor + 1 \\
 &= \left\lfloor \frac{200000-1}{86400} \right\rfloor + 1 = \left\lfloor \frac{199999}{86400} \right\rfloor + 1 \\
 &= 2 + 1 = 3.
 \end{aligned}$$

Art-27. Pigeonhole Principle (Strong Form)

Let $q_1, q_2, \dots, q_n$ be positive integers. If

$$q_1 + q_2 + \dots + q_n - n + 1$$

objects are put into n boxes, then either the 1st box contains at least q_1 objects, the 2nd box contains at least q_2 objects, ..., the n th box contains at least q_n objects.

Proof. Suppose it is not true, that is, the i th box contains at most $q_i - 1$ objects, $i = 1, \dots, n$. Then the total number of objects contained in the n boxes can be at most

$$(q_1 - 1) + (q_2 - 1) + \dots + (q_n - 1) = q_1 + q_2 + \dots + q_n - n,$$

which is one less than the number of objects distributed. This is a contradiction.

The simple form of the pigeonhole principle is obtained from the strong form taking $q_1 = q_2 = \dots = q_n = 2$.

$$\text{Then } q_1 + q_2 + \dots + q_n - n + 1 = 2n - n + 1 = n + 1$$

Cor: In elementary mathematics the strong form of the pigeonhole principle is most often applied in the special case when $q_1 = q_2 = \dots = q_n = r$. In this case the principle becomes

- If $n(r - 1) + 1$ objects are put into n boxes, then at least one of the boxes contains r or more of the objects. Equivalently,
- If the average of n nonnegative integers $a_1, a_2, \dots, a_n$ is greater than $r - 1$, i.e.,

$$\frac{a_1 + a_2 + \dots + a_n}{n} > r - 1, \text{ then at least one of the integers is greater than or equal to } r.$$

Example 11. A basket of fruit is being arranged out of apples, bananas, and oranges. What is the smallest number of pieces of fruit that should be put in the basket in order to guarantee that either there are at least 8 apples or at least 6 bananas or at least 9 oranges?

$$\text{Sol. } 8 + 6 + 9 - 3 + 1 = 21$$

Example 12. Given two disks, one smaller than the other. Each disk is divided into 100 congruent sectors. In the larger disk 100 sectors are chosen arbitrarily and painted red. In the smaller disk each sector is painted either red or blue with no stipulation on the number of red and blue sectors. The smaller disk is placed on the larger disk so that the centers and sectors coincide. Show that it is possible to place the two disks so that the number of sectors of the smaller disk whose color matches the corresponding sector of the larger disk is at least 100.

Sol. We fix the larger disk first, then place the smaller disk on the top of the larger disk so that the centers and sectors coincide. There 200 ways to place the smaller disk in such a manner. For each such alignment, some sectors of the two disks may have the same color. Since each sector of the smaller disk will match the same color sector of the larger disk 100 times among all the 200 ways and there are 200 sectors in the smaller disk, the total number of matched color sectors among the 200 ways is $100 \times 200 = 20,000$. Note that there are only 200 ways. Then there is at least one way that the number of matched color sectors is $\frac{20,000}{200} = 100$ or more.

EXERCISE 3 (k)

1. Ten members of a club have Rs. 1001 in their pockets. Show that at least one of them must have at least Rs. 101 in his pocket.
2. How many people must you have to guarantee that at least 12 of them will have birthdays on the same day of the week ?
3. Show that if any five numbers from 1 to 8 are chosen, then two of them will add to 9.
4. Show that if any 30 people are selected, then one may choose a subset of five so that all five were born on same day of week.
5. Show that if 30 dictionaries in a library contain a total of 61,327 pages, then one of the dictionaries must have at least 2045 pages.
6. How many friends must you have to guarantee so that at least five of them will have birthday in the same month ?
7. Describe in brief the Pigeonhole Principle with example.

(Pbi.U., M.C.A., 2003)

8. State three forms of Pigeonhole Principle. Give an example of use of each form of the Pigeonhole principle.

(Pbi.U., M.Sc.I.T., 2008)

ANSWERS

2. 78

6. 49

Find the number of 4-combinations of $\{\infty, a_1, \infty, a_2, \infty, a_3, \infty, a_4, \infty, a_5\}$

Find the number of non-negative integral solution to $x_1 + x_2 + x_3 + x_4 + x_5 = 50$

How many integral solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$, where $x_i \geq 2 \quad \forall i = 1, 2, 3, 4, 5$

11. Find the number of integral solution to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$
where $x_1 \geq 3, x_2 \geq 2, x_3 \geq 4, x_4 \geq 6, x_5 \geq 0$

12. How many different outcomes are possible from tossing 10 similar dice?

13. Enumerate the number of non-negative integral solutions to the inequality

$$x_1 + x_2 + x_3 + x_4 + x_5 \geq 19$$

14. Find the number of integral solution to $x_1 + x_2 + x_3 + x_4 + x_5 \geq 50$
where

$$x_1 \geq 4, x_2 \geq 7, x_3 \geq -14, x_4 \geq 10$$

Answer

1.	(i) 1260 (ii) 1716	2.	(i) 45 (ii) 21 (iii) 35	3.	20	4.	54054
5.	(i) 720 (ii) 210 (iii) 360	6.	64				
7.	aaa, aab, aac, aad, bba, bbd, cca, ccb, ccd, abc, abd, acd, bcd			8.	70	9.	316, 251
10.	$C(14, 10)$	11.	$C(9, 5)$	12.	3003		
13.	$C(5-1+0, 4) + C(5-1+1, 4) + \dots + C(5-1+19, 4)$			14.	$C(54, 3)$		

14.5 Pigeonhole Principle

This principle is also known as **Dirichlet Drawer Principle** or the **Shoe Box Principle**. It can be stated as follows.

14.5.1 Theorem of Pigeonhole Principle [U.P.T.U. (B.Tech.) 2005, 2006, 2007; U.P.T.U. (M.C.A.) 2008]

If n pigeons are assigned to m pigeonholes, and $m < n$, then at least one pigeonhole contains two or more pigeons.

Proof: Let $H_1, H_2, \dots, H_m$ denote m pigeonholes and $P_1, P_2, \dots, P_m, P_{m+1}, \dots, P_n$ denote n pigeons. We consider

the assignment of n pigeons to m pigeonholes as follows:

Let the pigeon P_1 is assigned to pigeonhole H_1 , pigeon P_2 is assigned to pigeonhole H_2 ... and pigeon P_m to pigeonhole H_m . This assigns as many as pigeons possible to individuals pigeonholes. Since $m < n$, there are $(n-m)$ pigeons that have not yet been assigned. At least one pigeonhole will be assigned to a second pigeon.

Example 77: Show that among 13 people, there are at least two people who were born in the same month.

Solution: We consider 13 people as pigeons and 12 months (January, February, ... December) as the pigeonholes. Here the number of pigeons is greater than the number of pigeonholes, therefore by the pigeonhole principle there will be at least two people who were born in the same month.

Example 78: If eight people are chosen in any way from some group, at least two of them will have been born on the same day of the week. [Delhi (B.E.) 2005]

Solution: Here each person (pigeon) is assigned to the day of the week (pigeonhole) on which he or she was born. Since there are eight people and only seven days of the week, then by pigeon hole principle at least two people must be assigned to the same day of the week.

Example 79: Show that if we choose 9 single shoes out of 8 distinct pairs of shoes, then we are sure to have a pair. [Rohtak (B.E.) 2008]

Solution: Here 8 distinct pairs correspond to 8 pigeonholes and 9 single shoes correspond to pigeons. It means $m = 8$ and $n = 9$, $m < n$ then by pigeonhole principle, there must be one pigeonhole with 2 shoes which means that we have chosen at least one pair.

Example 80: Find the minimum number of students in a class so that three of them are born in the same month. [Kurukshetra (B.E.) 2007; Osmania (B.E.) 2009]

Solution: Here $n = 12$ (months) which represent the number of pigeonholes.

$$m = k + 1 = 3 \text{ or } k = 3, k \text{ is positive integer}$$

Hence, the minimum number of pigeons $= kn + 1 = 2 \times 12 + 1 = 25$ students.

Example 81: How many students must there be in a class to guarantee that at least two students receive the same score in the final examination if the examination is graded on a scale from zero to one hundred points.

Solution: Here $m = 2$ But $m = k + 1$, k is any integer

then

$$k + 1 = 2 \Rightarrow k = 1$$

$$m = 101$$

So, total number of students in class to guarantee that at least two students receive the same score.

$$= kn + 2 = 1 \times 101 + 1 = 102.$$

Example 82: Show that if any five integers from 1 to 8 are chosen, then at least two of them will have a sum 9.

Solution: Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$

The different sets, each containing two numbers whose sum is equal to 9 are

$$A_1 = \{1, 8\}, A_2 = \{2, 7\}, A_3 = \{3, 6\}, A_4 = \{4, 5\}.$$

Each of the five numbers chosen from 1 to 8 must belong to one of these sets. The four sets are considered as pigeonholes and five chosen numbers are taken as pigeons. Since $m = 4$, $n = 5$ i.e. $4 < 5$ Then by pigeonhole principle we can say that the two of the selected numbers must belong to the same set whose sum is 9.

Example 83: Show that in any room of people who have been doing handshaking there will always be at least two people who have shaken hands the same number of times.

Solution: Let the pigeonholes be labelled with the different numbers of hands shaken and we put the people (pigeons) into their correct pigeonhole. Suppose there are n people, then since only shake hands with each person at most once, the labels on the pigeonholes will go from 0 to $(n-1)$ i.e. we have n holes and n people. It is not possible for the 0^{th} and $(n-1)^{\text{th}}$ the holes both to be occupied, because if one person has shaken hands with nobody then there can not be any one person who has shaken hands with every other person. Thus, we have at most $(n-1)$ holes occupied at any one time. Hence, by the pigeonholes principle at least one of the holes has two occupants, which shows that there are at least two people who have shaken hands the same number of time.

Example 84: What shall be the minimum number of words that must begin with the same alphabet at 27 English words.

Solution: Here, $m = kn + 1 = 27$ (Pigeonholes), $n = 26$ (number of alphabet) = (pigeons) - 1, where k is any integer

$$\therefore k \times 26 + 1 = 27 \Rightarrow k = 1 \quad \text{and} \quad m = k + 1 = 2$$

Hence at least two words must begin with the same alphabet.

Example 85: If there are 13 boys in a class, then find minimum number of boys born in the same months.

Solution: Consider month as pigeons i.e. $n = 12$ then find pigeonholes m . But $n = 12$, $kn + 1 = 13$

$$\Rightarrow k \times 12 + 1 = 13 \Rightarrow k = 1, \quad \text{then} \quad m = k + 1 = 2$$

Example 86: Show that if seven numbers from 1 to 12 are chosen, then two of them will add up to 13.

Solution: There are six sets each containing two number that add up to 13 as follows:

$$A_1 = \{1, 12\}, A_2 = \{2, 11\}, A_3 = \{3, 10\}, A_4 = \{4, 9\}, A_5 = \{5, 8\}, A_6 = \{6, 7\}$$

Each of the seven numbers belongs to one of these six sets. Since there are 6 sets, then by pigeonholes principle two of the numbers chosen belong to the same set. These numbers add upto 13.

Example 87: If 20 candidates appear in a competitive examination then show that there exists at least two among them whose Roll number differ by a multiple of 19. [U.P.T.U. (B.Tech.) 2004]

Solution: Let X be the set of Roll number of 20 candidates and Y be the set of remainder when any positive integer is divided by 19.

$$\text{i.e.,} \quad Y = \{0, 1, 2, \dots, 18\}.$$

Here we define a function $f: X \rightarrow Y$ as $f(x)$ = the remainder obtained when any positive integer is divided by 19. Since $|X| = 20$ and $|Y| = 19$ i.e., $|X| > |Y|$, therefore by the pigeonhole principle function f cannot be one-one. This shows that there exists two distinct Roll numbers x_1 and x_2 such that $f(x_1) = f(x_2)$

$$x_1, x_2 \text{ can be written as } x_1 = 19n_1 + f(x_1) \text{ and } x_2 = 19n_2 + f(x_2)$$

where n_1 and n_2 are positive integers.

$$x_1 - x_2 = 19(n_1 - n_2)$$

$$[\because f(x_1) = f(x_2)]$$

this gives

Now $n_1 - n_2$ is an integer. Therefore $x_1 - x_2$ is a multiple of 19.

Hence there are at least two candidate whose Roll numbers differ by a multiple of 19.

Suppose, if possible, that there is no increasing or decreasing subsequence of length $n + 1$ in the sequence a_i of $n^2 + 1$ integers. Then it is clear that the set $\{(x_k, y_k)\}$ has at most n^2 distinct ordered pairs (pigeonholes). It follows by the pigeonhole principle that there must exist two elements a_i and a_j in the sequence which correspond to the same ordered pair; i.e., $(x_i, y_i) = (x_j, y_j)$. But this is not possible because if $a_i < a_j$ then we must have $x_i > x_j$ and if $a_i > a_j$ then we must have $y_i > y_j$. In each case, we have a contradiction. This shows that there is either an increasing subsequence or a decreasing subsequence of length $n + 1$.

14.5.2 The Extended Pigeonhole Principle

Theorem: If n pigeons are assigned to m pigeonholes, then one of the pigeonholes must contain at least $\left\lceil \frac{n-1}{m} \right\rceil + 1$ or $\left\lfloor \frac{n}{m} \right\rfloor + 1$ pigeons. $\left\lfloor \frac{p}{q} \right\rfloor$ denotes the largest integer, less than or equal to the rational number $\frac{p}{q}$.

Proof: We shall prove this theorem by contradiction. Let each pigeonhole contain no more than $\left\lfloor \frac{n-m}{m} \right\rfloor$

pigeons. Then, there are at most $m \cdot \left\lfloor \frac{n-1}{m} \right\rfloor$ pigeons in all but $\left\lfloor \frac{n-1}{m} \right\rfloor < \frac{m(n-1)}{m} \Rightarrow \left\lfloor \frac{n-1}{m} \right\rfloor = n-1$

i.e. there $(n-1)$ pigeons in all. Which is contradiction our assumption. Hence one of the pigeonholes must contain at least $\left\lfloor \frac{n-1}{m} \right\rfloor + 1$ pigeons.

Example 92: Show that if any 26 people are selected then we may choose a subset of 4 so that all 4 were born on the same day of the week.

Solution: We assign each person to the day of the week on which she or he was born. Then number of pigeons (people) are to be assigned to 7 pigeonholes with $n = 26$ and $m = 7$ at least.

$$\left\lfloor \frac{n-1}{m} \right\rfloor + 1 \text{ or } \left\lceil \frac{n}{m} \right\rceil + 1 = \left\lfloor \frac{26-1}{7} \right\rfloor + 1 = 4 \text{ people}$$

must have been on the same day of the week.

Example 93: There are 38 different time periods during which classes at university can be scheduled if there are 677 different classes, how many different rooms will be needed?

Solution: Assign each class to a time period then 677 different classes (pigeons) has to be assigned to 38 pigeonholes. Here $n = 677$, $m = 38$.

Then by generalized pigeonhole principle, one of the pigeonholes must contain at least

$$\left\lfloor \frac{677}{38} \right\rfloor + 1 = [17.81] + 1 = 17 + 1 = 18 \text{ pigeons}$$

Example 94: In a group of six people, each pair of individuals consists of two friends or two enemies. Show that there are either three mutual friends or three mutual enemies in the groups.

Solution: Let A be one of the six people of group. There are either three or more among the other five people who are friends of A or enemies of A . Then by generalized pigeonhole principle five objects are divided into two sets, one has at least $\left\lfloor \frac{5}{2} \right\rfloor + 1 = 3$ objects. Let B, C and D are friends of A . If any two of these individual are friends then these two and A form a group of three mutual friends. Otherwise B, C and D form a set of three mutual enemies. In the second case suppose B, C and D are enemies of A . If any two of these individual are enemies then these two and A form a group of three mutual enemies otherwise B, C and D form a group of three mutual friends.

Example 95: How many letters one must choose from the set of 15 A's, 20 B's and 25 C's, so that 12 identical letters will always be included in the selection.

Solution: We see that there are three types of letters (pigeonholes) to be chosen from the set having 15 A's, 20 B's and 25 C's, so that 12 identical letters should always be included in the selection is the smallest integer n such that

$$k+1=12,$$

or

$$\left\lfloor \frac{n-1}{3} \right\rfloor + 1 = 12 \Rightarrow [n-1] = 11 \times 3 \Rightarrow n = 34$$

Thus, 34 is the minimum number of letters to be chosen from the set having 15 A's, 20 B's and 25 C's so that 12 identical letters will always be included in the selection.

Example 96: What is the minimum number of students required in a class to be that at least five will receive the same grade if there are four possible grades A, B, C and D.

Solution: There are four grade A, B, C, D which is the pigeonholes and at least 5 students must receive the same grade. We have to find minimum number of students which is pigeons (n) such that

$$\left\lfloor \frac{n-1}{4} \right\rfloor + 1 = 5$$

or

$$\left\lfloor \frac{n-1}{4} \right\rfloor = 4 \text{ or } (n-1) = 16 \text{ or } n = 17.$$

Example 97: Show that if seven colours are used to paint 50 cars, at least eight cars will have the same colour.

Solution: Let $n = 50$ cars = pigeons, $m = 7$ colours = pigeonholes

Then, by generalized pigeonhole principle $\left\lfloor \frac{n-1}{m} \right\rfloor + 1 = \left\lfloor \frac{50-1}{7} \right\rfloor + 1 = 8$ Cars

will be the same colour.

Example 98: Show that among 1000 people there are at least 84 people who were born in the same month.

Solution: Where $n = 1000$ people = pigeons, $m = 12$ months = pigeonholes

Then, by generalized pigeonhole principle, we have $\left\lfloor \frac{n-1}{m} \right\rfloor + 1 = \left\lfloor \frac{1000-1}{12} \right\rfloor + 1 = 83 + 1 = 84$ people.

who were born in the same month.

Example 99: Prove that among 1,00,000 people there are two who born on the same time.

[R.G..P.V. (B.E.) Bhopal 2001, 2007]

Solution: Let $n =$ pigeons = 1,00,000 people.

A person can be born at any second = $24 \times 60 \times 60 = 86,400$ seconds i.e., $m = 86,400 =$ pigeonholes. By generalized pigeonhole principle

$$\left\lfloor \frac{1,00,000-1}{86,400} \right\rfloor + 1 = (1.16) + 1 = 1 + 1 = 2 \text{ people are born on same time.}$$

Exercise

¶ Pigeonhole

1. State and prove pigeonhole Principle.

[U.P.T.U. (B.Tech.) 2007; U.P.T.U. (M.C.A.) 2008; R.G.P.V. (B.E.) Bhopal 2001, 2002, 2003, 2009;

2. If a bag contains several white, red and black shoes, then find the minimum number of shoes that a person needs to choose in order to get two pairs (4) shoes of the same colour.

Rohatak (B.E.) 2007; P.T.U. (B.E.) 2005; M.K.U. (B.E.) 2005, 2009]

[Hint: $n = 4, k + 1 = m = 5 \Rightarrow kn + 1 = 17$]

3. If there are several black, white and green balls in a bag, then find the number of balls that must be chosen to be sure of getting pair of balls with the same colour.
4. A professor tells three stories from 'Panchtantra' in his class each year. How many stories does the professor require in order to never repeat exactly the same triple of stories over a period of 12 years.
5. If there are 104 different pairs of people who know each other at a party of 30 people, show that some person has at most acquaintances.
6. Given any ten points inside an equilateral triangle of side 1, show that there are two points within a distance of at most $1/3$.
7. How many persons must be chosen in order that atleast 7 of them will have birthdays in the same calender month?

[Hint: Divide the given triangle into 9 square triangle]

[Hint: $\left\lceil \frac{n-1}{12} \right\rceil + 1 = 7$]

8. How many letters one must choose from the set of 15 A's 20 B's and 25 C's, so that 12 identical letters will always be included in the selection.
9. Show that atleast in any of the seven classes there must be two that meet on the same day that no class hold on Sunday.
10. 11 shoes are selected from 10 pairs of shoes. Show that there is a pair of matched shoes among the selection.
11. Show that in a set of five distinct integers, there must exists two integers with the same remainder when divided by 4.
12. Show that if any five numbers from 1 to 8 are chosen then two of them will add upto 9.
13. Show that if any 30 people are selected, then we may choose a subset of 5 so that all 5 were born on the same day of the week.
14. If 30 processors are inter connected, show that atleast two processors are directly connected to the same number of processors.
15. There are four doors to enter a hall and three doors to leave the hall. In how many ways can a person enter and leave the hall?
16. Explain the pigeon hole principle.

[U.P.T.U. (M.C.A.) 2009]